

Herbivorous insects independently evolved salivary effectors to regulate plant immunity by destabilizing the malectin-LRR RLP NtRLP4

Xin Wang, Jia-Bao Lu, Yi-Zhe Wang, Xu-Hong Zhou, Jian-Ping Chen, Chuan-Xi Zhang, Jun-Min Li , Hai-Jian Huang 

State Key Laboratory for Quality and Safety of Agro-Products, Key Laboratory of Biotechnology in Plant Protection of MARA, Zhejiang Key Laboratory of Green Plant Protection, Institute of Plant Virology, Ningbo University, Ningbo, China • Institute of Applied Ecology, Fujian Agriculture and Forestry University, Fuzhou, China • Office of Science and Technology, Yunnan University of Chinese Medicine, Kunming, China

 https://en.wikipedia.org/wiki/Open_access

 Copyright information

eLife Assessment

This study provides an **important** contribution by showing that whiteflies and planthoppers use salivary effectors to suppress plant immunity through the receptor-like protein RLP4, suggesting convergent evolution in these insect lineages. The topic is of clear interest for understanding plant-insect interactions and offers ideas that could stimulate further research in the field. However, the strength of evidence is **incomplete**, as some aspects of the data and experimental design limit the extent to which the main claims are fully supported.

<https://doi.org/10.7554/eLife.108737.1.sa3>

Abstract

Plants rely on receptor-like proteins and receptor-like kinases (RLPs/RLKs) to detect a variety of invading pathogens and insects. However, it remains unclear how these invaders are able to overcome receptor perception and disable the plant signaling pathways. Here, we show that whiteflies and planthoppers independently evolve salivary effectors to target defensive RLP4, therefore minimizing the plant immunity. RLP4 is a leucine-rich repeat RLP (LRR-RLP) that confers plant resistance against herbivorous insects by forming RLP4/SOBIR1 complexes. In whitefly *Bemisia tabaci*, BtRDP is an Aleyrodidae-restricted protein that attaches to salivary sheath. It interacts with RLP4 from different plants, causing the degradation of RLP4 in an ubiquitin-dependent manner. Transgenic plants overexpressing NtRLP4 exerts a detrimental impact on *B. tabaci* by exploiting the crosstalk between the salicylic acid (SA) and jasmonic acid (JA) pathways. Conversely, the overexpression of BtRDP or silencing of NtRLP4 effectively alleviates such negative effects. In planthopper *Nilaparvata lugens*, a Delphacidae-restricted salivary protein NISP694 also targets and promotes degradation of OsRLP4 from rice plants. Our results reveal the convergent evolution of salivary proteins in insects, and shed lights on the complex interactions between plants and herbivorous insects.

Introduction

The activation of innate immunity by membrane-localized receptors is a highly conserved mechanism in eukaryotes [1]. In sessile plants, the first line of defense against potential pathogenic microbes or insects relies on cell surface pattern-recognition receptors (PRRs) [2]. The plant genome encodes hundreds of PRRs, including receptor-like kinases (RLKs) and receptor-like proteins (RLPs) [2]. RLKs consist of different types of N-terminal ectodomains, a single transmembrane domain, and a cytoplasmic protein kinase domain, while RLPs have a similar structure but lack the kinase domain [3, 4]. As RLPs lack intracellular signaling domains, they are usually associated with SOBIR1 or SOBIR1-like adaptors to form bimolecular receptor kinases [4]. Despite the large number of PRRs encoded by plant genomes, only a few of them have been well characterized in terms of their function in recognizing microbial-associated molecular patterns (MAMPs), damage-associated molecular patterns (DAMPs), and herbivore-associated molecular patterns (HAMPs) [3, 5]. For instance, FLS2 is known to recognize bacterial flagellin [6], EFR can recognize bacterial elongation factor Tu [7], and EIX2 is responsible for recognizing the fungal ethylene-inducing xylanase [8].

Piercing-sucking insects, such as whiteflies, aphids, and planthoppers, damage plants by feeding or transmitting viruses. They inject a mixture of saliva into plant tissues, which aids their performance by binding to the leaked calcium, degrading extracellular DNA, and regulating plant hormonal signals [9–12]. In response, plants have evolved the ability to detect herbivores by sensing HAMPs present in insect saliva [5, 13]. This triggers pattern-triggered immunity (PTI), including mitogen-activated protein kinase (MAPK) cascades, reactive oxygen species (ROS), and hormone signaling [14–16]. Although a few salivary proteins have been reported to activate plant PTI, the specific PRRs that recognize these proteins and trigger plant immunity are still not well understood [17–19]. One exception is the inceptin receptor (INR), which is a legume-specific LRR-RLP recognizing the Vu-In in insect oral secretions [5]. Moreover, plant PRRs have been widely reported to be targeted by pathogen effectors [20–22]. It remains unknown how insects cope with the immune recognition system.

The whitefly *Bemisia tabaci* (Hemiptera: Aleyrodidae) and brown planthopper *Nilaparvata lugens* (Hemiptera: Delphacidae) are notorious pests globally. During the feeding process, both insects secrete abundant species-specific salivary proteins into host plants [23, 24]. The plants have been reported to initiate defense responses when they recognize insect feeding [17, 25]. However, the specific PRRs responsible for this process are unknown. Both insects can successfully ingest phloem saps with limited plant defenses. It remains unclear how insects attenuate plant defenses. Given that many defense mechanisms are conserved across plants [1, 26], it is interesting to investigate whether insect evolves a similar strategy to aid feeding in convergent evolution. In this study, we characterize the function of RLPs in plant immunity responsive to piercing-sucking insects, and demonstrate that *B. tabaci* and *N. lugens* independently evolve salivary proteins to regulate the posttranslational modification of RLPs.

Results

BtRDP is most abundantly expressed in salivary glands and secreted into host plants

According to the transcriptomic results, BtRDP (*B. tabaci* RLP-degrading protein; GenBank accession: MN738093) is the most abundantly expressed gene in salivary glands [24]. It is a secretory protein containing a signal peptide at the N-terminal, and totally 6 unique peptides from

BtRDP are detected in *B. tabaci* watery saliva (Fig. S1). As there is no gene homolog identified in the NCBI nr database, BtRDP is subjected to BLAST analysis against the transcriptomic or genomic data of 30 insect species. The results show that RDP is distributed in all the analyzed Aleyrodidae species, while no RDP homolog is detected in other species (Table S1). Based on phylogenetic analysis, RDPs from five *B. tabaci* cryptic species are clustered in the same clade, showing a close relationship to RDP from *Aleurocanthus spiniferus* (Fig. S2).

BtRDP can be secreted into host plants, with two bands between 25-35kDa being detected in *Nicotiana tabacum* plants infested by *B. tabaci*, but not in the non-infested plants (Fig. 1a). The smaller high intensity band is the same size as that of BtRDP in salivary glands, and always presents in the *B. tabaci*-infested plants. For the larger lower intensity band, it presents in infested samples with lower frequency. It is possible that the larger band might be the BtRDP that undergoes modification after being secreted into plants, similar with another salivary effector previously reported [27]. However, we cannot exclude the possibility that this larger band is the non-specific *N. tabacum* protein induced by *B. tabaci* feeding. Western-blotting, qPCR and transcriptomic analysis demonstrate that BtRDP is nearly exclusively expressed in the salivary glands (Fig. 1b, c; Fig. S3a), which is significantly different from the actin or ribosomal 18S rRNA (Fig. S4a). In addition, BtRDP is mainly expressed in the adult stages, but lowly expressed in the egg, nymph, and pseudopupa stages (Fig. 1d; Fig. S3b), similar to another salivary protein BtFTSP [28] but different from actin or ribosomal 18S rRNA (Fig. S4b). Immunohistochemical (IHC) staining shows that BtRDP is specifically distributed in principle salivary gland, but not in the accessory salivary gland or other tissues (Fig. 1e). A few salivary proteins have been reported to attach to salivary sheath after secretion [29]. We also stain the salivary sheath secreted by *B. tabaci* using anti-BtRDP serum. The results show that BtRDP signal is mainly detected in the protuberant structure of sheath (Fig. S5), indicating that BtRDP attaches to salivary sheath after secretion.

BtRDP is important for *B. tabaci* feeding on tobacco plants

RNA interference is firstly employed to investigate the potential role of BtRDP in whitefly feeding, and dsBtRDP treatment efficiently and specifically suppresses the target gene at transcriptional and protein level (Fig. 2a; Fig. S6). We do not observe a significant difference in salivary sheath morphology between dsBtRDP- and dsGFP-treated *B. tabaci* (Fig. S7a). However, the salivary sheath length of dsBtRDP-treated *B. tabaci* is significantly shorter than that of the dsGFP-treated control (Fig. S7b). Moreover, dsBtRDP treatment exerts no influence on the whitefly survivorship (Fig. S7c). For reproduction analysis, the dsBtRDP-treated whiteflies oviposit less eggs than the control (Fig. 2b). The whitefly feeding behavior, which can be classified into nonpenetration (np), pathway duration (C), phloem salivation (E1), and phloem ingestion (E2) [30] (Fig. S8), is recorded using the electrical penetration graph (EPG) technique. Compared with dsGFP treatment, dsBtRDP-treated *B. tabaci* exhibits a dramatic decrease in phloem ingestion (Fig. 2c). Moreover, dsBtRDP-treated whitefly exhibits a significantly increased pathway duration phase and a slightly increased nonpenetration phase (Fig. 2c).

Salivary proteins are reported to function in intercellular space and/or extracellular space after secreted into host plants [29, 31]. To investigate the role of BtRDP in different subcellular location of host plants, we construct two transgenic *N. tabacum* overexpressing BtRDP, one with a complete coding region (oeBtRDP) and the other without signal peptide (oeBtRDP^{sp}). The signal peptide of BtRDP may enable protein secretion into the extracellular space. The flag tag is fused to the C-terminal ends of recombinant proteins, respectively. The empty vector (EV) transgenic plants, which exert no significant influence on insect performance compared with the WT (Fig. S9), is used as a negative control. Fecundity assay demonstrates that both oeBtRDP and oeBtRDP^{sp} transgenic plants are beneficial to whiteflies, with more eggs oviposited in these two overexpressing plants compared with the EV controls (Fig. 2d-i; Fig. S10). Two-choice

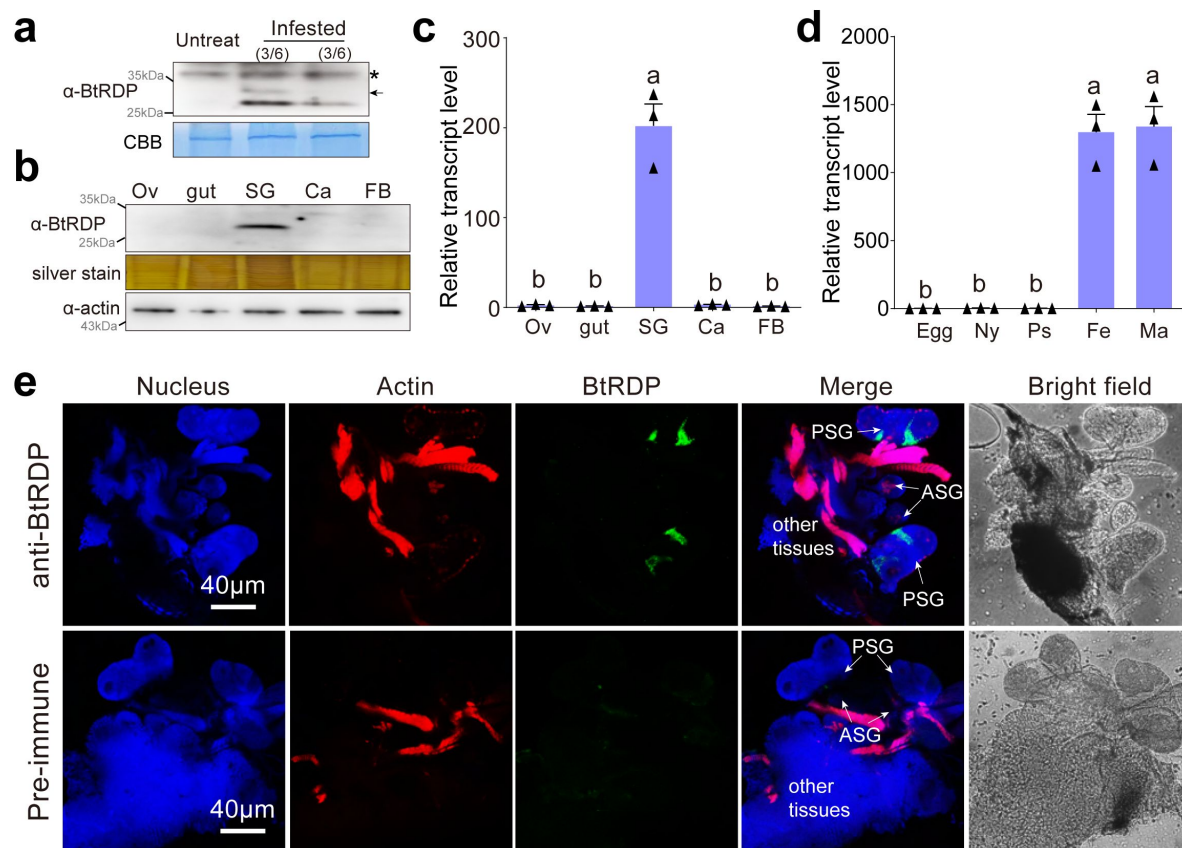


Fig. 1.

BtRDP is a salivary protein and secreted into plants.

(a) Detection of BtRDP in *Nicotiana tabacum* plants. The untreated and *Bemisia tabaci*-infested tobacco plants are collected for western-blotting assays. Six independent *B. tabaci*-infested plants are tested. The presence of lower intensity band (arrow) in each replicate is summarized in the bracket. Asterisk indicates non-specific binding. Coomassie brilliant blue (CBB) staining is conducted to visualize the sample loading. (b, c) Detection of BtRDP in different tissues by western-blotting and qRT-PCR assays. Ov, ovary; SG, salivary gland; Ca, carcass; FB, fat body. Silver staining and anti-actin serum are used to visualize the sample loading. (d) Expression patterns of BtRDP in different development stages. Ny, nymph; Ps, pseudopupa; Fe, female; Ma, male. *B. tabaci tubulin* is used as an internal control. The relative quantitative method ($2^{-\Delta\Delta Ct}$) is used to evaluate the quantitative variation. Data are presented as mean values $\pm$ SEM ($n=3$ independent biological replicates). Different lowercase letters indicate statistically significant differences at the $P < 0.05$ level according to one-way ANOVA test followed by Tukey's multiple comparisons test. (e) Immunohistochemical staining of BtRDP. The salivary gland and its nearby tissues are dissected and incubate with anti-BtRDP serum or pre-immune serum conjugated with Alexa Fluor™ 488 NHS Ester (green) and actin dye phalloidinrhodamine (red). The nucleus is stained with DAPI (blue). PSG, principal salivary gland; ASG, accessory salivary gland. Experiments are repeated thrice for (b), while twice for (e). Similar results are observed and representative images are displayed.

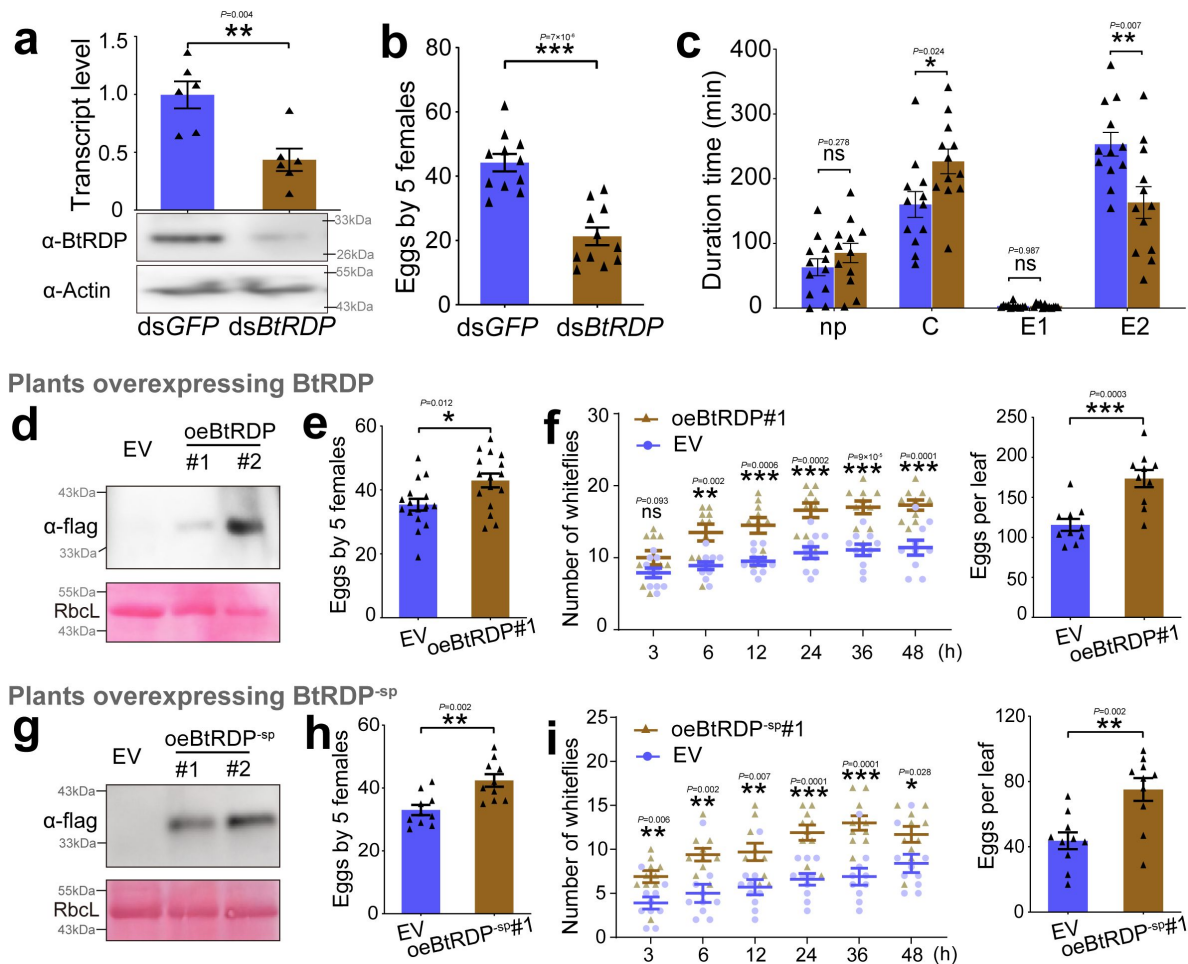


Fig. 2.

Effects of BtRDP on *Bemisia tabaci*.

(a) Treating *B. tabaci* with dsBtRDP significantly reduces the transcript and protein level of target gene. The dsGFP-treated *B. tabaci* is used as a control. qRT-PCR data are presented as mean values $\pm$ SEM ($n=6$ independent biological replicates). (b-c) Effects of BtRDP knockdown on insect reproduction (b) and feeding behavior (c). Electrical penetration graph (EPG) is used to monitor the insect feeding behavior, which can be classified into nonpenetration (np), pathway duration (C), phloem salivation (E1) and phloem ingestion (E2) phases. All EPG recordings are performed for 8 h. Typical EPG waveforms are displayed in Fig. S8. (d-f) Insect performance on transgenic plants overexpressing BtRDP (oeBtRDP). (d) Detection of BtRDP in transgenic *Nicotiana tabacum* overexpressing a complete coding region of BtRDP. The empty vector (EV) plants are used as the control. The flag tag is fused to the C-terminal ends of recombinant proteins. (e) Comparison of insect reproduction on oeBtRDP#1 and EV plants. Five *B. tabaci* individuals are confined to indicated plants for 3 days, and the oviposited eggs are counted. (f) Attraction of oeBtRDP#1 and EV leaves to *B. tabaci* in a two-choice equipment. A group of 40 female *B. tabaci* are released into a device containing oeBtRDP#1 and EV leaves. The number of insects settling on each leaf is counted. After 48 h, the number of eggs on each leaf is counted. (g-i) Insect performance on transgenic plants overexpressing BtRDP^{sp} (oeBtRDP^{sp}). (g) Detection of BtRDP^{sp} in transgenic *N. tabacum* overexpressing BtRDP without a signal peptide. The insect reproduction (h) and settlement (i) on oeBtRDP^{sp} transgenic plants are recorded. Data are presented as mean values $\pm$ SEM. The EPG data are first checked for normality and homogeneity of variance, and data not fitting a normal distribution are subjected to log10 transformation. *P*-values are determined by two-tailed unpaired Student's *t* test. ****P* < 0.001; ***P* < 0.01; **P* < 0.05; ns, not significant. Western-blotting assays are repeated thrice with the similar results.

experiments exert similar results, with more insects being settled and oviposited in oeBtRDP or oeBtRDP^{SP} transgenic plants than the EV controls (**Fig. 2d-i**; Fig. S10). These results indicate that BtRDP is beneficial to whiteflies, regardless of its subcellular location in host plants.

Transcriptomic sequencing is performed on EV, oeBtRDP#1, and oeBtRDP^{SP}#1 transgenic plants. PCA analysis demonstrates that the biological replicates of each transgenic plant can be well clustered (Fig. S11a). Totally 1253 and 1227 differentially expressed genes (DEGs) are identified when comparing oeBtRDP#1 and oeBtRDP^{SP}#1 with EV plants, respectively (Table S2; Table S3). Enrichment analysis of DEGs in these two comparison groups exerts similar results, with pathways associated with MAPK signaling, environmental adaptation, signal transduction, and plant-pathogen interaction significantly enriched (Fig. S10b, c). There are 464 DEGs overlap in these two comparison groups (Fig. S11d). Interestingly, the majority of these DEGs exhibit similar expression pattern in oeBtRDP#1 and oeBtRDP^{SP}#1 when compared with EV plants (Fig. S11e), suggesting that BtRDP affect some plant pathways regardless of its subcellular location. There are 1084 DEGs identified in comparison of oeBtRDP#1 and oeBtRDP^{SP}#1 plants (Table S4). Enrichment analysis show that the DEGs associated with metabolism, diterpenoid biosynthesis, and plant-pathogen interaction are significantly enriched (Fig. S12), suggesting that subcellular location of BtRDP also has non-negligible influence on host plants.

BtRDP interacts with NtRLP4/NtSOBIR1 complex

To investigate the underlying mechanism of BtRDP in improving whitefly feeding, yeast two-hybrid (Y2H) screening against a *N. benthamiana* cDNA library is conducted using BtRDP^{SP} as a bait. Totally seven proteins are found to potentially interact with BtRDP^{SP} (Table S5), including an NbRLP4 (accession Niben261Chr07g1310001.1 in Niben261 genome). NbRLP4 contains a predicted N-terminal signal peptide, a malectin-like domain, a LRR domain, followed by a transmembrane domain. NbRLP4, as well as its closest homology in *N. tabacum* (NtRLP4; GenBank accession: XM_016601109), does not contain the kinase domain (**Fig. 3a**).

The relative expression of *NtRLP4* in response to *B. tabaci* infestation is investigated. qRT-PCR results demonstrate that the transcript level of *NtRLP4* is not significantly influenced by *B. tabaci* (Fig. S13). However, a decrease in protein level of NtRLP4 is detected by western-blotting assay (**Fig. 3b**). Point-to-point Y2H assays reveal that NtRLP4 without signal peptides and transmembrane domains interacts with BtRDP^{SP} (**Fig. 3c**). To investigate the specificity of NtRLP4-BtRDP interaction, another RLP protein NtCf9 showing the highest sequence similarity to tomato resistance protein Cf9 [32], as well as three randomly selected salivary proteins from *B. tabaci*, are used as controls. The results show that BtRDP^{SP} cannot interact with truncated NtCf9 without signal peptides and transmembrane domains (**Fig. 3c**). Also, NtRLP4₍₂₃₋₅₄₁₎ cannot interact with the other three salivary proteins without signal peptides (Fig. S14a, b). In addition, it is the LRR domain, but not the malectin-like domain or the short intercellular region, that is required for the NtRLP4-BtRDP interaction (**Fig. 3c**). To investigate the shortest LRR sequence responsible for BtRDP interaction, LRR domain is divided into three segments. However, none of these segments interact with BtRDP^{SP} (Fig. S14a). Co-immunoprecipitation (Co-IP) assays further confirm the specific interaction between BtRDP and NtRLP4 (**Fig. 3d**). It is the C-terminal myc tag-fused NtRLP4 (NtRLP4-myc), but not the NtCf9 (NtCf9-myc), that can be immunoprecipitated by C-terminal flag tag fused BtRDP (BtRDP-flag). In bimolecular fluorescence complementation (BiFC) assay, YFP fluorescence is observed when co-expressing the N-terminal nYFP tag fused NtRLP4 and N-terminal cYFP tag fused BtRDP^{SP} (Fig. S14c). Furthermore, oeBtRDP transgenic plants that overexpressing BtRDP-flag are incubated with anti-flag beads. The results show that endogenous NtRLP4 can be immunoprecipitated, which is significantly different from the control (Fig. S14d). These results suggest that BtRDP interacts with NtRLP4.

The absence of kinase motifs in the C-terminal region of NtRLP4 suggests that it must function in cooperation with other proteins to initiate the signaling. The receptor-like kinase SOBIR1, which contained a kinase domain (Fig. S15a), is reported to be required for the function of several RLPs

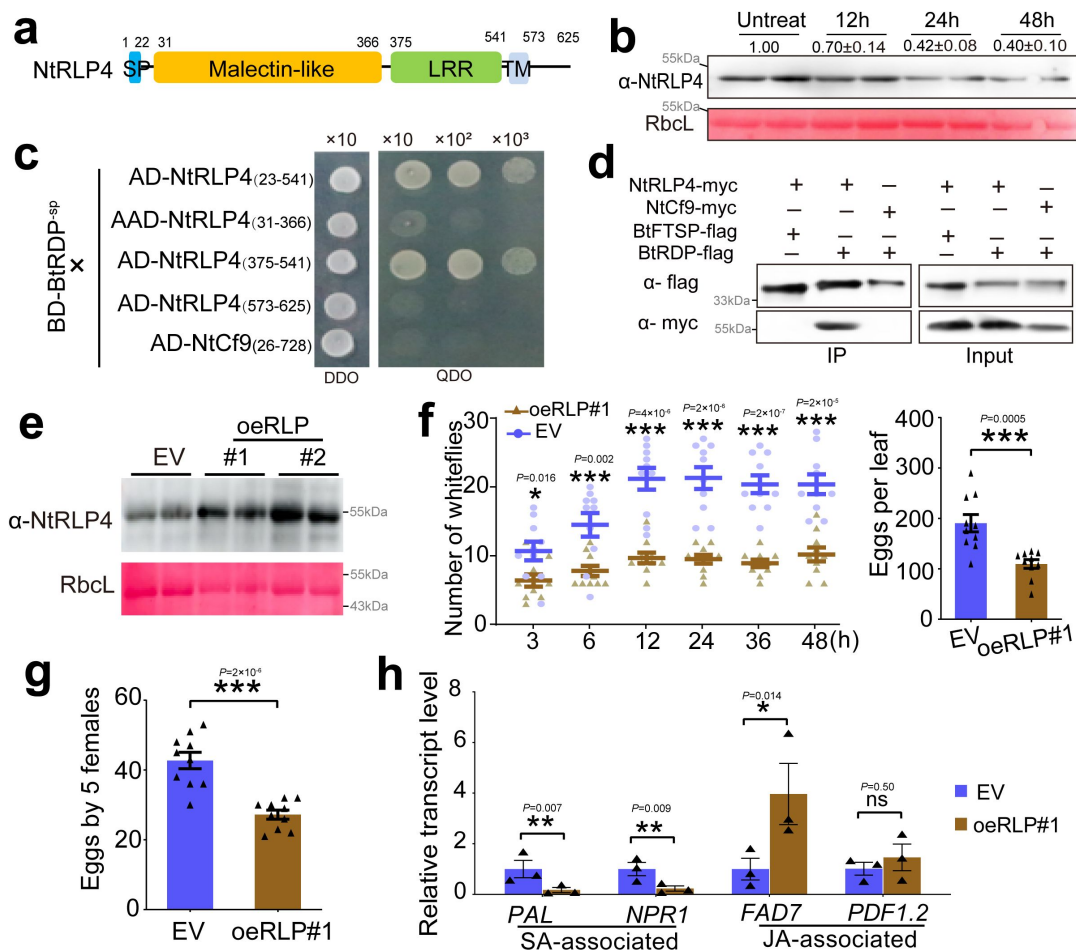


Fig. 3.

NtRPL4 interacts with BtRDP and confers plant resistance to *Bemisia tabaci*.

(a) Domain organization of NtRPL4. NtRPL4 contains a predicted N-terminal signal peptide (SP), a malectin-like domain, a LRR domain, and a transmembrane domain (TM). **(b)** The protein level of NtRPL4 in response to *B. tabaci* infestation. The untreated *Nicotiana tabacum* is used as a negative control. Four independent biological replicates are performed, and the representative images are displayed. The band density is measured using ImageJ. The density values from 4 biological replicates are calculated and the mean value in the controls is set at 1.0. **(c)** Yeast two hybrid assays showing the interaction between BtRDP and NtRPL4. The different combinations of constructs are transformed into yeast cells, and are grown on the selective medium SD/-Trp/-Leu (DDO), and the interactions are tested with SD/-Trp/-Leu/-His/-Ade (QDO). **(d)** Co-immunoprecipitation assay showing the interaction between BtRDP and NtRPL4. Total proteins are extracted from *N. benthamiana* leaves transiently co-expressing NtRPL4-myc/NtCf9-myc with BtFTSP-flag/BtRDP-flag. All genes are expressed with a complete coding region, and the myc/flag tags are fused at the C-terminus. Precipitation is performed using flag beads. The samples are probed with anti-flag and anti-myc antibodies for immunoblotting analysis. **(e-h)** Analysis of transgenic *N. tabacum* overexpressing NtRPL4 (oeRPL). **(e)** Detection of NtRPL4 level in oeRPL plants. The empty vector (EV) plant is used as a control. Two independent oeRPL lines are selected. The samples are probed with an anti-NtRPL4 antibody. Rubisco staining (Rbcl) is used to visualize the amount of sample loading. **(f)** Attraction of oeRPL#1 and EV leaves to *B. tabaci* in a two-choice equipment. A group of 40 female *B. tabaci* are released into a device containing oeRPL#1 and EV leaves. The number of insects settling on each leaf is counted at each time point. After 48 h, the number of eggs on each leaf is counted. **(g)** Comparison of insect reproduction on oeRPL#1 and EV plants. Five *B. tabaci* individuals are confined to indicated plants for 3 days, and the oviposited eggs are counted. **(h)** Relative transcript levels of salicylic acid (SA)- and jasmonic acid (JA)-associated genes in oeRPL#1 and EV plants. PAL, phenylalanine ammonia lyase; NPR1, nonexpressor of pathogenesis-related protein 1; FAD7, fatty acid desaturase 7; PDF1.2, plant defensin 1.2. Two independent biological replicates are performed in **(d)** and **(e)**. Data in **(f)**, **(g)**, and **(h)** are presented as mean values ± SEM. For insect bioassays in **(f)** and **(g)**, *n* = 10 independent biological replicates. For qRT-PCR in **(h)**, *n* = 3 independent biological replicates. *P*-values are determined by two-tailed unpaired Student's *t* test. ****P* < 0.001; ***P* < 0.01; **P* < 0.05; ns, not significant.

involved in the innate immunity [4]. Through Co-IP assays, our study demonstrates that NtRPL4 is capable of interacting with NtSOBIR1 (Fig. S15b).

NtRPL4 and NtSOBIR1 are resistant to *B. tabaci* infestation

The transgenic *N. tabacum* overexpressing NtRPL4 (oeRPL#1 and oeRPL#2, without any tags) is constructed (Fig. 3e). Compared with the EV control, both oeRPL lines are less attractive to *B. tabaci*, with less settling and ovipositing insects (Fig. 3f; Fig. S16a). Also, we observe the significantly decreased fecundity of *B. tabaci* in oeRPL plants (Fig. 3g; Fig. S16b). Similar results are observed after transient overexpression of NtRPL4-GFP in *N. tabacum* (Fig. S16c, d), indicating that NtRPL4 is resistant to *B. tabaci* infestation.

Thereafter, the EV and oeRPL#1 plants are subjected to transcriptomic sequencing (Fig. 17a). Totally 729 differentially expressed genes (DEGs) are identified, including 423 up-regulated and 306 down-regulated ones (Table S6). Enrichment analysis demonstrate that the majority of up-regulated DEGs are associated with plant-pathogen interaction, environmental adaptation, MAPK signaling pathway, and signal transduction (Fig. 17b). By contrast, the glutathione metabolism, carbohydrate metabolism, and amino acid metabolism pathways are significantly enriched in down-regulated DEGs (Fig. 17c). Noteworthy, there are many DEGs annotated as RLK/RLP or WRKY transcription factor, with the majority of them being significantly upregulated (Fig. S17d, e). These results suggest an increased defense in oeRPL plants. The crosstalk between salicylic acid (SA) and jasmonic acid (JA) pathways is critical for plant defenses [12]. The increased defense of oeRPL transgenic plant is partially caused by the altered hormonal signaling. The JA-associated genes *FAD7* is significantly induced, while the SA-associated genes *PAL* and *NPR1* are repressed in oeRPL plants (Fig. 3h). Transient overexpression of NtRPL4-GFP in *N. tabacum* plants via *Agrobacterium* infiltration exert similar results (Fig. S16e). Furthermore, NtRPL4-GFP significantly enhances the H₂O₂ accumulation (Fig. S16f).

We also investigate the function of NtSOBIR1 by transient overexpression and hairpin RNAi (Fig. S18). Overexpression of NtSOBIR1-GFP induces the cell death phenotype five days post agro-injection (Fig. S18b). Insects prefer to settle and oviposit on GFP-expressed *N. tabacum* than on NtSOBIR1-GFP-expressed control, even no obvious death phenotype is observed two days post agro-injection (Fig. S18c, d). For NtSOBIR1 silencing, no cell death phenotype is observed, while the NtSOBIR1-silenced leaves exhibit reduced growth (Fig. S18f). Compared with EV-silenced control, NtSOBIR1-silenced *N. tabacum* is slightly attractive to *B. tabaci* (Fig. S18g, h). Collectively, these results demonstrate that NtSOBIR1 is resistant to *B. tabaci* infestation.

BtRDP attenuates plant defenses by promoting the degradation of host RLP4

To investigate the interplay between NtRPL4 and BtRDP, we construct transgenic *N. tabacum* silencing NtRPL4 (*RNAi-RLP#1* and *RNAi-RLP#2*) (Fig. S19a). Compared with EV plants, *B. tabaci* show a tendency to settle on NtRPL4-silenced plants and lay more eggs (Fig. S19b, c). Thereafter, *dsGFP*- and *dsBtRDP*-treated *B. tabaci* are reared on EV, *RNAi-RLP#1*, and *RNAi-RLP#2*, respectively. The *dsBtRDP*-treated *B. tabaci* produce less offspring than the control (decreased by 28.3%; $P=0.0007$) on EV plants (Fig. 4a), consistent with the results on WT plants (Fig. 2b). On *RNAi-RLP#1* plant, the impair insect reproduction is still observed after *dsBtRDP* treatment, but the extent is slightly reduced (decreased by 23.7%; $P=0.002$) (Fig. 4a). On *RNAi-RLP#2* plant, there is no statistical significance between *dsGFP* and *dsBtRDP* treatments (Fig. 4a). These results suggest that *RNAi-RLP* plants are beneficial for insects and can partially, but not completely, rescue the impaired feeding performance of *dsBtRDP*-treated insects.

To investigate the influence of BtRDP on NtRPL4, the GFP-tagged NtRPL4 (NtRPL4-GFP) is transiently co-expressed with mCherry-tagged BtRDP (BtRDP-mCherry), red fluorescent protein (RFP)-mCherry, and BtFTSP1-mCherry via agroinfiltration, respectively. All tags are fused to the C-

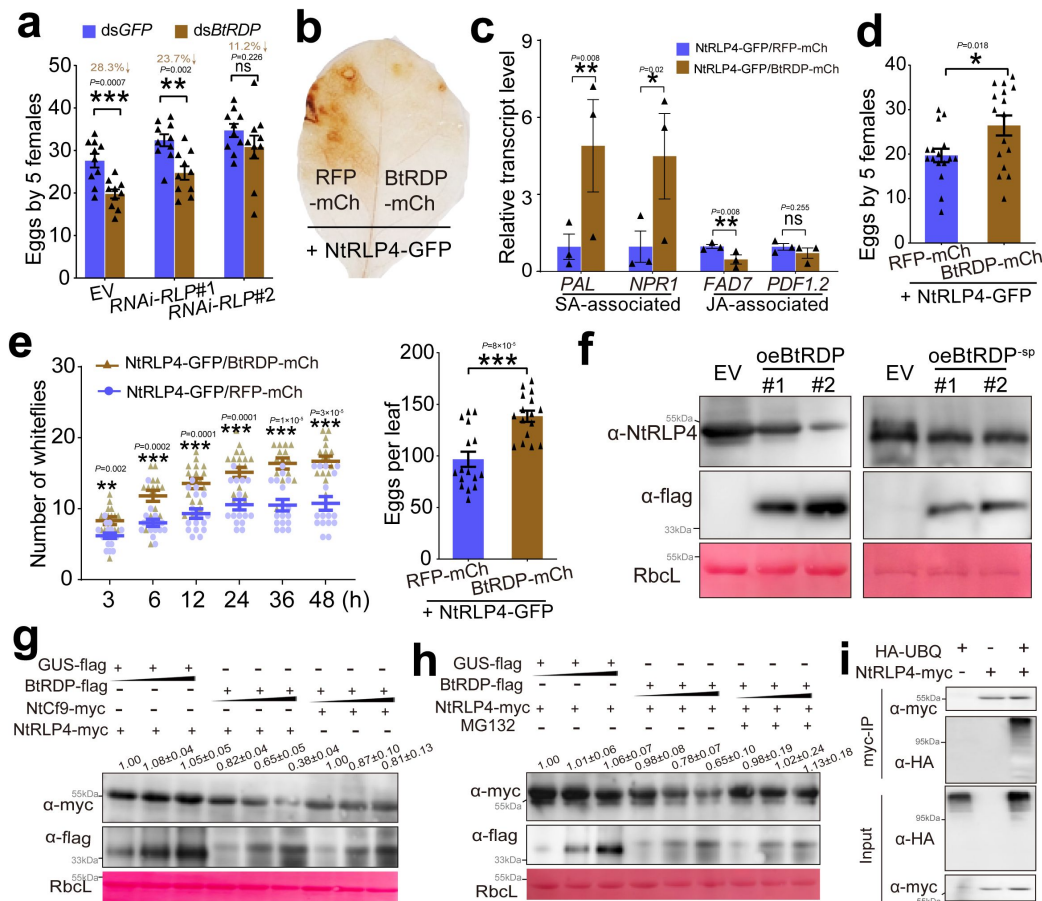


Fig. 4.

BtRDP suppress plant defenses by promoting NtRLP4 degradation.

(a) Effects of dsBtRDP suppression on insect reproduction ($n = 10$ independent biological replicates) when feeding on empty vector (EV) and NtRLP4-silenced (RNAi-RLP) transgenic *Nicotiana tabacum*. **(b)** Transient overexpressing BtRDP-mCherry attenuates H₂O₂ accumulation caused by NtRLP4-GFP overexpression. *N. tabacum* co-expressing RFP-mCherry and NtRLP4-GFP is used as a control. All genes are expressed with a complete coding region. The experiment is repeated 5 times with the similar results. **(c)** Relative transcript level of salicylic acid (SA)- and jasmonic acid (JA)-associated genes in BtRDP-mCherry/NtRLP4-GFP and RFP-mCherry/NtRLP4-GFP plants ($n = 3$ independent biological replicates). PAL, phenylalanine ammonia lyase; NPR1, nonexpressor of pathogenesis-related protein 1; FAD7, fatty acid desaturase 7; PDF1.2, plant defensin 1.2. **(d)** Comparison of insect reproduction ($n = 16$ independent biological replicates) on transient-expressed BtRDP-mCherry/NtRLP4-GFP and RFP-mCherry/NtRLP4-GFP plants. **(e)** Attraction of BtRDP-mCherry/NtRLP4-GFP and RFP-mCherry/NtRLP4-GFP leaves to *B. tabaci* in a two-choice equipment ($n = 16$ independent biological replicates). The number of eggs on each leaf is counted at 48 h post insect release. **(f)** Effects of BtRDP and BtRDP^{sp} overexpression on NtRLP4 accumulation. Transgenic oeBtRDP (left) and oeBtRDP^{sp} (right) plants are probed with anti-NtRLP4 and anti-flag antibodies for immunoblotting analysis. The experiments are repeated twice with the similar results. **(g)** Degradation of NtRLP4 by BtRDP in *N. benthamiana* leaves. NtRLP4-myc and BtRDP-flag are transiently co-expressed in *N. benthamiana* plants through *Agrobacterium* infiltration. *Agrobacterium* carrying NtCf9-myc and GUS-flag are used as negative controls. **(h)** Effects of 26S proteasome inhibitor (MG132) on NtRLP4 accumulation. Co-infiltrated leaves are treated with MG132 at 24 h post injection. The samples are probed with anti-flag and anti-myc antibodies for immunoblot analysis. Rubisco staining (RbcL) is conducted to visualize the amount of sample loading. The small triangle indicates the different concentrations of *Agrobacterium* (OD₆₀₀ = 0.05, 0.3, and 1.0). **(i)** NtRLP4 is ubiquitinated *in planta*. NtRLP4-myc is co-expressed transiently with HA-UBQ in *N. benthamiana* leaves. Extracted total proteins are immunoprecipitated by anti-myc beads and immunoblotted with anti-myc or anti-HA antibody. Experiments are repeated thrice for **(g)** and **(h)**, while twice for **(i)**. Band density is measured using ImageJ. The density values from three biological replicates are calculated and the mean value in the first lane is set at 1.0. Data are presented as mean values $\pm$ SEM. P -values are determined by two-tailed unpaired Student's t test. *** $P < 0.001$; ** $P < 0.01$; * $P < 0.05$; ns, not significant.

terminus. BtRDP-mCherry does not influence the location patterns of NtRPL4-GFP (Fig. S20). However, a significant reduction in the fluorescent signal of NtRPL4-GFP is detected when co-expressed with BtRDP-mCherry, but not the controls (Fig. S20). The co-expression of BtRDP-mCherry/NtRPL4-GFP significantly reduces H₂O₂ accumulation (Fig. 4b) and suppresses the expression of the JA-associated genes *FAD7* and *PDF.12*, while inducing the expression of the SA-associated genes *PAL* and *NPR1* (Fig. 4c). Insect bioassays demonstrate that transient overexpression of BtRDP-mCherry rescues the negative effects on whitefly performance caused by NtRPL4-GFP overexpression (Fig. 4d, e). In host choice experiments, *B. tabaci* display a preference for settling on *N. tabacum* leaves co-expressing BtRDP-mCherry/NtRPL4-GFP over those co-expressing RFP-mCherry/NtRPL4-GFP (Fig. 4d). We observe almost no preference in *B. tabaci* feeding on *N. tabacum* plants co-expressing BtRDP-mCherry/NtRPL4-GFP and plants expressing GFP alone (Fig. S21).

The attenuated NtRPL4-GFP fluorescent signal (Fig. S20), along with the observed decrease in the protein level of NtRPL4 during *B. tabaci* infestation (Fig. 3b), leads us to speculate on whether BtRDP influences the accumulation of NtRPL4. To this end, the BtRDP^{SP}-his and GFP-his proteins are prokaryotic expressed in *Escherichia coli*. Different concentrations of purified BtRDP^{SP}-his and GFP-his proteins are infiltrated into *N. tabacum* plants overexpressing NtRPL4-myc. Interestingly, the protein level of NtRPL4-myc shows a slight decrease with increasing amounts of BtRDP^{SP}-his, while no such effect is observed with GFP-his (Fig. S22). Moreover, the BtRDP^{SP}-his protein does not have significant effect on the accumulation of another RLP NtCf9-like (Fig. S22).

Then, we detect the NtRPL4 level in oeBtRDP and oeBtRDP^{SP} plants. Compared with the EV control, a reduced NtRPL4 abundance is observed in both oeBtRDP#1 and oeBtRDP#2 transgenic lines, but not the oeBtRDP^{SP} plants (Fig. 4f), suggesting the importance of BtRDP secretion in extracellular space. BtRDP attaches to salivary sheath that localizes in plant apoplast (Fig. S5). The presence of signal peptide promotes the secretion of BtRDP into extracellular space. Therefore, we use the complete coding region of *BtRDP* in subsequent experiments. The NtRPL4-myc and BtRDP-flag are transiently co-expressed in *N. benthamiana* plants through *Agrobacterium* infiltration. The results show that BtRDP-flag significantly decreasing the protein level of NtRPL4-myc, but not the NtCf9-myc (Fig. 4g). qRT-PCR analyses demonstrate that the transcript level of *NtRPL4* remains unchanged (Fig. S23). These findings indicate that the reduction in protein level of NtRPL4 is not caused by transcriptional changes but rather to degradation.

The ubiquitin system and autophagy are two major pathways regulating protein degradation [33]. At first, we employ MG132, a 26S proteasome inhibitor known to inhibit the degradation of ubiquitin-conjugated proteins [34]. The results show that MG132 efficiently prevents the degradation of NtRPL4-myc by BtRDP-flag (Fig. 4h). Then, we employ autophagy inhibitor 3-Methyladenine (3-MA) and bafilomycin A1 (BAF) that can block autophagy in plants [35]. However, the decreased protein level of NtRPL4-myc is still observed in the presence of 3-MA or BAF (Fig. S24). Additionally, NtRPL4-myc and HA-tagged ubiquitin (HA-UBQ) are co-expressed in *N. benthamiana*. Total proteins are extracted from *N. benthamiana* leaves and then subjected to immunoprecipitation with anti-myc beads. The ubiquitin attached to NtRPL4 is determined by immunoblot using anti-HA antibody. Leaves expressing NtRPL4-myc or UBQ-HA alone are used as negative controls. The results show that a strong UBQ signal is detected in the immunoprecipitated product only when NtRPL4-myc and UBQ-HA are co-expressed (Fig. 4i). The molecular weight of UBQ-attached NtRPL4-myc is remarkably larger than NtRPL4-myc itself (Fig. 4i). These results suggest that BtRDP promotes NtRPL4 degradation through the ubiquitin system, but not the autophagy pathway.

Degrading plant RLP4s by salivary effectors is a common strategy for insects

To investigate the ability of BtRDP to degrade RLPs from other host plants of *B. tabaci*, *Solanum lycopersicum* RLP4 (SlRLP4; GenBank accession: XP_004232910) is selected for analysis. Sequence analysis reveals a similar gene structure between SlRLP4 and NtRLP4 (Fig. S25; Fig. S26a). The interaction between BtRDP and SlRLP4 is confirmed through Y2H and Co-IP assays (Fig. S26b, c). Moreover, we transiently co-express SlRLP4-myc and SlCf9-like-myc with varying concentrations of BtRDP-flag or GUS-flag, with the myc and flag tags fused to the C-terminus. The results show that BtRDP-flag exerts no influence on the accumulation of SlCf9-like-myc protein. By contrast, the protein level of SlRLP4-myc is significantly reduced in the presence of BtRDP-flag (Fig. S26d), indicating the conserved function of BtRDP in *B. tabaci* adaptation to different host plants.

BtRDP is an Aleyrodidae-restricted gene that does not have homologues in other species. To explore whether RLP degradation by salivary effectors is conserved among different herbivorous insects, a RLP4 homology from *Oryza sativa* (OsRLP4; GenBank accession: XP_015645303; Fig. 5a) and twenty salivary proteins from planthopper species without signal peptides are paired to identify the potential interactions using Y2H assays. As a result, only the salivary protein NlSP694^{SP} (GenBank accession: MF278694) interacts with OsRLP4₍₂₉₋₅₅₁₎ (OsRLP4 without signal peptides and transmembrane domains), while the other salivary proteins analyzed do not (Fig. 5b, c; Fig. S27). NlSP694 does not show any sequence or structure similarity to BtRDP (Fig. S28). Based on truncation analysis, NlSP694^{SP} interacts with both the LRR domain and malectin-like domain of OsRLP4 (Fig. 5c), slightly different from that of the BtRDP-NtRLP4 interaction (Fig. 3c). Considering that both BtRDP and NlSP694 interact with the LRR domain of RLP4, the specific region of LRR domain responsible for NlSP694 binding is further investigated. The LRR domain of OsRLP4 is divided into three segments, in accordance with the LRR domain division in NtRLP4 (Fig. S14a). NlSP694^{SP} interacts with the truncated LRR domain (OsRLP4₍₄₈₀₋₅₂₆₎) (Fig. 5c). By contrast, none of the truncated LRR domain of NtRLP4 interact with BtRDP^{SP} (Fig. S14a). These results suggest that the binding sites of salivary proteins to RLP4 may be different between *N. lugens* and *B. tabaci*.

NlSP694 is a planthopper-specific salivary protein that is nearly exclusively expressed in salivary glands (Fig. 5d; Fig. S29). The function of NlSP694 or its gene homologs in other planthopper species has not been well investigated previously. Silencing NlSP694 does not impact insect survivorship (Fig. 5e). However, significant reductions in honeydew excretion and impaired fecundity are observed in dsNlSP694-treated *N. lugens* (Fig. 5f, g), indicating the importance of NlSP694 in insect feeding.

Subsequently, we transiently co-express OsRLP4-myc with a complete coding region of NlSP694 (C-terminal flag tagged, NlSP694-flag). Another salivary protein NlSP7-flag and NtCf9-myc are used as negative controls. The results demonstrate that NlSP694-flag specifically and significantly decreases the protein level of OsRLP4-myc, which differs significantly from that of the NlSP7-flag (Fig. 5h). Additionally, we demonstrate that NlSP694-flag fails to decrease the protein level of NtCf9-myc (Fig. 5h). These findings suggest that *N. lugens* potentially evolves salivary effectors to target rice RLPs.

Discussion

Plants have evolved a complex innate immune system to protect themselves against pathogen/insect invasion, and the cell surface-localized PRRs function on the frontline [2]. To counteract this response, the pathogens have been reported to develop various methods to avoid plant PRR-mediated PTI for their own advantage [36]. However, it remains unknown whether

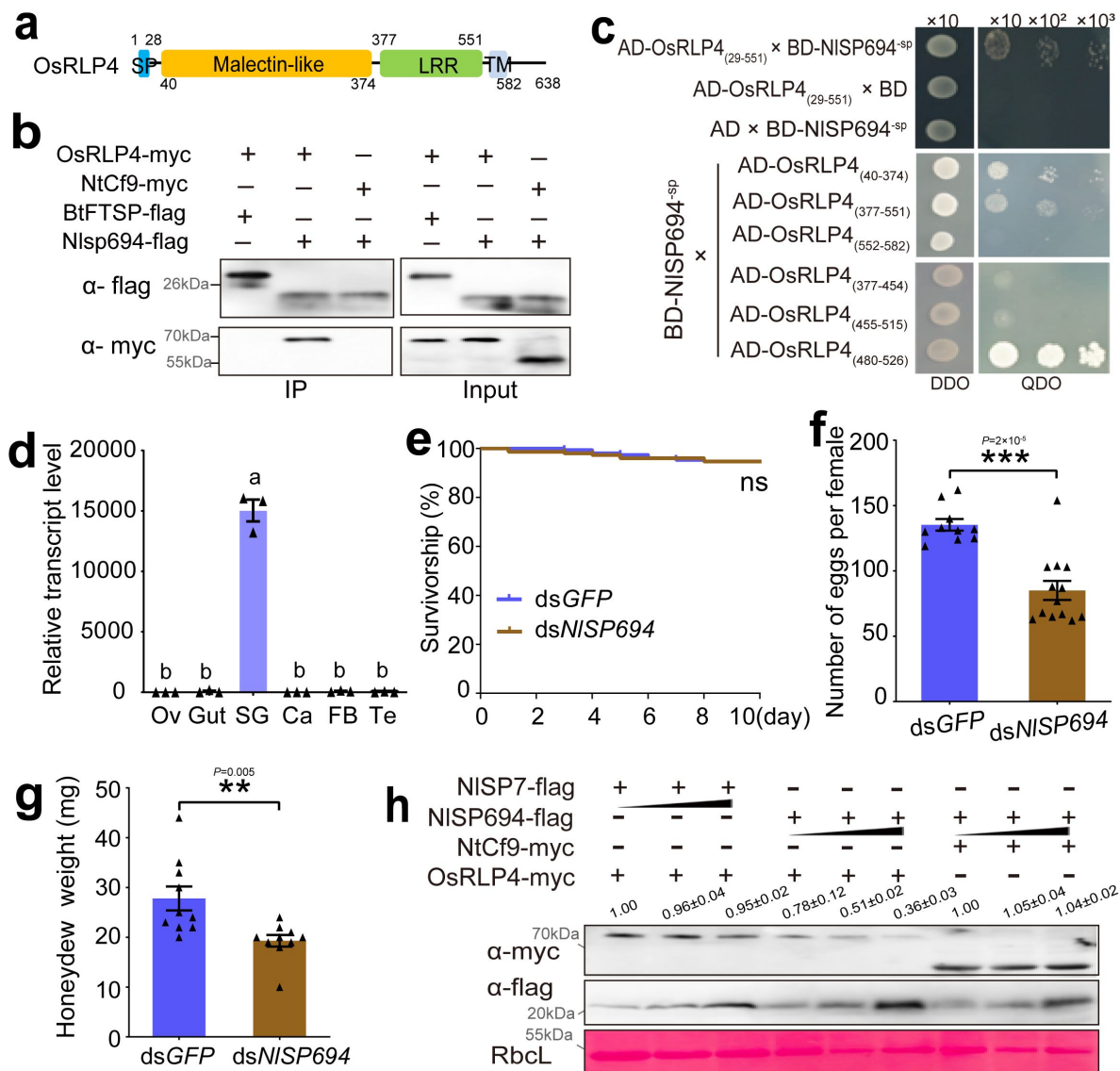


Fig. 5.

Rice RLP4 are targeted by salivary protein NISP694 from *Nilaparvata lugens*.

(a) Domain organization of *Oryza sativa* RLP4 (OsRLP4). (b, c) Yeast two-hybrid and Co-IP assays showing the interaction between OsRLP4 and NISP694. All genes are expressed with a complete coding region in Co-IP assays, while NISP694 without a signal peptide is used in yeast two-hybrid assays. Experiments in (b) are repeated twice with the similar result. (d) The expression patterns of NISP694 in different *N. lugens* tissues. Te, testis; Ov, ovary; SG, salivary gland; Ca, carcass; FB, fat body. Data are presented as mean values $\pm$ SEM ($n = 3$ independent biological replicates). Different lowercase letters indicate statistically significant differences at $P < 0.05$ level according to one-way ANOVA test followed by Tukey's multiple comparisons test. (e-g) Effects of dsRNA treatment on insect survivorship (e), reproduction (f), and honeydew excretion (g). For survivorship analysis, a group of 30 *N. lugens* are reared in a cage. Three independent biological replications are performed. Differences in survivorship between the two treatments are tested by log-rank test. ns, not significant. For reproduction analysis, $n = 10$ and 13 individuals are tested in dsGFP- and dsNISP694-treatment, respectively. The sterile females are excluded from data analysis. For honeydew analysis, $n = 10$ independent biological replicates. P -values in (f) and (g) are determined by two-tailed unpaired Student's t test. *** $P < 0.001$; ** $P < 0.01$. (h) Effect of NISP694 on the accumulation of transient-expressed OsRLP4. OsRLP4-myc is agro-injected into *Nicotiana benthamiana* together with different concentrations of NISP694-flag and NISP7-flag. NtCf9-myc is used as a negative control. The small triangle indicates the different concentrations ($OD_{600} = 0.05, 0.3$, and 1.0) of *Agrobacterium*. Rubisco staining (RbcL) is conducted to visualize the amount of sample loading. Experiments are repeated three times with the similar results. Band density is measured using ImageJ. The density values from three biological replicates are calculated with the mean value in the first lane being set at 1.0.

insects take the similar strategy to sabotage plant immunity. In this study, we demonstrate that the RLP4/SOBIR1 complexes confer plant resistance against herbivorous insects. In response, both planthoppers and whiteflies secrete salivary effectors that facilitate the degradation of defensive RLP4, leading to attenuated plant immunity. Our results suggest that manipulation of PRRs may be a conserved strategy for herbivorous insects.

PRRs recognize modified-self or non-self patterns and activate a series of immune signaling, which subsequently regulate downstream defense responses and confer plant resistance against invaders [2, 3]. While various insect salivary components have been reported to trigger plant immune responses, the specific PRRs responsible for this process are elusive. Our study demonstrates that RLP4 exerts a significant impact on insect performance by modulating plant hormones and ROS (Fig. 3f-h; Fig. S16-S17), highlighting the critical role of this gene in plant immunity. Nevertheless, the exact ligand for RLP4 remains unknown. It is possible that malectin-LRR receptor RLP4 responds to the DAMPs or HAMPs that are released during insect feeding [37], which deserved further investigation. RLP4 lacks the intracellular kinase domain, and forms bimolecular receptor kinases by interacting with SOBIR1 (Fig. S15). Overexpression of NtRLP4 or NtSOBIR1 enhances insect feeding, while silencing of either gene exerts the opposite effect (Fig. Fig. S16-S19). The association between RLPs and SOBIR1 protein kinases is a structural and functional equivalent to genuine receptor kinases [4]. Therefore, disrupting the stability of the RLP4/SOBIR1 complex can potentially undermine the immune recognition process.

As pivotal regulators in immune signaling, PRRs are frequently targeted by multiple pathogens through diverse mechanism. For example, *Pseudomonas syringae* hampered the functionality of receptor complexes by disrupting the ligand-dependent association between receptor-like kinase BAK1 and the flagellin receptor FLS2 [22], while *Phytophthora infestans* fostered alternative splicing of the host RLPs to enhance infection [21]. PRRs are usually subjected to degradation via post-translational modifications [38]. In plants, the activity and stability of PRRs undergo stringent regulation since excessive activation of PRRs can be detrimental [20, 39]. One prevalent regulatory mechanism involves ubiquitination followed by subsequent degradation mediated by the 26S proteasome [38]. Mounting evidence has demonstrated the exploitation of such mechanisms by bacteria, fungi, viruses, and oomycetes to facilitate PRR degradation [40–43]. Nevertheless, the occurrence of effector-mediated PRR degradation in insects is unreported. Our study reveals that the salivary effectors BtRDP from *B. tabaci* and NISP694 from *N. lugens* interact with plant RLP4s, leading to their degradation in an ubiquitin-dependent manner (Fig. 4f-i). Notably, BtRDP and NISP694 share no sequence or structural similarity and lack resemblance to known eukaryotic ubiquitin-ligase domains. Their interaction with RLP4s occurs in the extracellular space (Fig. 3d; Fig. 5c), whereas the ubiquitin-proteasome system primarily functions in the cytosol and nucleus [44]. Furthermore, NtRLP4 reduction is observed only in oeBtRDP transgenic plants, not in oeBtRDP^{sp} plants (Fig. 4f), suggesting that BtRDP exerts its influence on NtRLP4 in the extracellular space. These findings might exclude the possibility that BtRDP/NISP694 harbors E3 ligase activity that directly ubiquitinates RLP4s in plant cells. It is likely that salivary proteins indirectly influence the post-translational modification of RLP4, increasing its degradation rate, which warrants further investigation.

Herbivorous insects use a combination of salivary components to facilitate successful feeding [29]. Previous studies have shown that *B. tabaci* suppresses plant defenses by leveraging SA-JA crosstalk, with the salivary protein Bt56 playing a pivotal role during this process [12]. Bt56 triggers the SA-signaling pathway by directly interacting with a KNOTTED 1-like homeobox transcription factor [12]. This study investigates the effects of another salivary effector, BtRDP, on plant hormones, and finds that it elicits similar effects to Bt56, albeit primarily impacting the signal recognition process (Fig. 3h; Fig. S16e). This leads us to speculate that different salivary effectors from the same insect species might simultaneously target the same signaling pathway to ensure effective immune suppression. Our analysis reveals that BtRDP interacts with other host genes in addition to NtRLP4 (Table S5), and silencing NtRLP4 cannot completely rescue the

impaired feeding performance of ds*BtRDP*-treated insects (**Fig. 4a**). Salivary proteins exert a crucial role in mediating the interactions between herbivores and plants. The dynamic interplay between plants and insects might be highly complex, one salivary protein can target multiple plant pathways, and one plant pathway is influenced by multiple salivary proteins.

In conclusion, this study shows that suppressing PRR-mediated plant immunity may be the common strategy for the successful feeding of herbivorous insects. Whiteflies and planthoppers independently evolve salivary effectors that facilitate the degradation of to degrade defensive RLP4 of host plants in an ubiquitin-dependent manner, which can therefore disrupt the RLP4/SOBIR1 complexes for signal recognition (**Fig. 6**).

Material and methods

Insects and plants

A colony of *B. tabaci* (cryptic species MED) was originally collected from a soybean field in Suzhou, Anhui Province, China. The insects were subsequently maintained on *N. tabacum* cv. K326 plants under controlled laboratory conditions ($25 \pm 1^\circ\text{C}$, 50–70% relative humidity, 16:8 h light:dark photoperiod). For experimental use, *N. benthamiana* and *N. tabacum* plants were grown in a climate-controlled growth chamber ($23 \pm 1^\circ\text{C}$, 16:8 h light: dark cycle).

Phylogenetic analysis of RDP

To identify potential RDP homologs across insect species, we performed a BLASTp search using the BtRDP protein sequence as a query against the predicted proteomes of 30 insect species. Hits were filtered using an E-value cutoff of 10^{-5} . The retrieved RDP homologs were aligned using MAFFT (v7.310) with default parameters [45]. To improve alignment quality, we removed ambiguously aligned regions using Gblocks with default settings. The optimal amino acid substitution model (JTT+G) was determined using ModelTest-NG (v0.1.6). Phylogenetic relationships were inferred using maximum likelihood analysis in RAxML (v0.9.0) with 1000 bootstrap replicates to assess branch support [46]. The final tree was visualized and annotated using iTOL (<https://itol.embl.de/>).

Preparation of anti-BtRDP and anti-NtRLP4 serum

The coding sequence of BtRDP (excluding the signal peptide) was PCR-amplified and cloned into the pET-28a vector (Novagen) to generate a C-terminal 6×His-tagged fusion protein. The recombinant plasmid was transformed into *Escherichia coli* Transetta (DE3) competent cells (TransGen Biotech, Beijing, China). Protein expression was induced with 0.5 mM isopropyl-β-D-thiogalactoside (IPTG) at 28°C for 8 h. Bacterial cells were harvested by centrifugation and lysed via ultrasonication for 30 min. The recombinant BtRDP^{SP}-His protein was purified under native conditions using Ni-NTA agarose (Qiagen, Hilden, Germany) according to the manufacturer's protocol.

Purified BtRDP^{SP}-His protein was used to immunize rabbits (Hua'an Biotechnology, Hangzhou, China) for custom polyclonal antibody production. The anti-NtRLP4 serum, prepared by immunizing rabbits with peptide GQNSDRPISTKNST, was produced via the custom service of Genscript Biotechnology Company (Nanjing, China). The anti-BtRDP serum was subsequently conjugated to Alexa Fluor™ 488 NHS Ester (#A20000, Thermo Fisher Scientific) following standard labeling protocols.

The specificity of anti-BtRDP serum was tested by comparing ds*GFP*- and ds*BtRDP*-treated *Bemisia tabaci*. The results showed that ds*BtRDP* significantly reduced the target gene. Similarly, the specificity of anti-NtRLP4 serum was tested by comparing RNAi-EV and RNAi-NtRLP4 plants, which

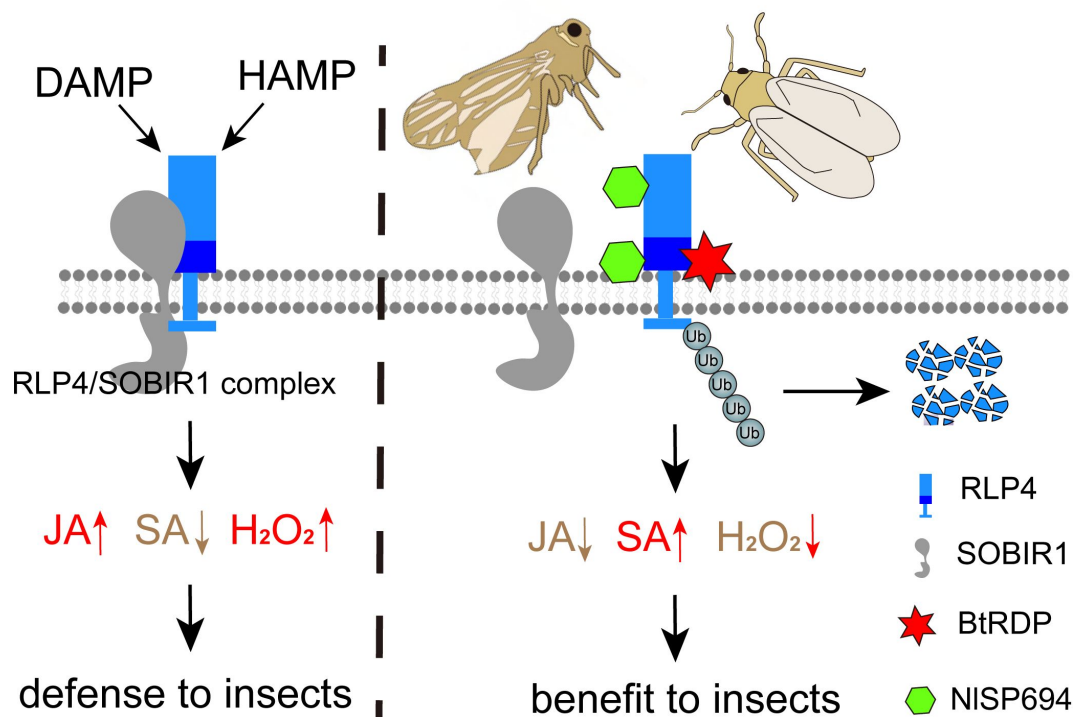


Fig. 6.

The proposed model for the suppression of receptor-like proteins (RLPs)-mediated plant defenses by salivary effectors.

Host plants employ pattern-recognition receptors (PRRs) to detect various damage-associated molecular patterns (DAMPs) and herbivore-associated molecular patterns (HAMPs) triggered by insect feeding. The RLP4/SOBIR1 complex plays a vital role in initiating pattern-triggered immunity (PTI), including H_2O_2 burst, upregulation of jasmonic acid (JA), and downregulation of salicylic acid (SA), which hinders insect feeding. The whitefly *Bemisia tabaci* and planthopper *Nilaparvata lugens* independently evolved salivary proteins that targeted plant RLP4. *B. tabaci* salivary sheath protein BtRDP interacts with the leucine-rich repeat (LRR) domain of RLP4 from *Nicotiana tabaci* and *Solanum lycopersicum*, while *N. lugens* NISP694 targets both the LRR domain and the malectin-like domain of *Oryza sativa* RLP4. These interactions promote the ubiquitin-dependent degradation of RLP4, thereby disrupting the stability of the RLP4/SOBIR1 complex. The presence of salivary effectors causes a hormonal shift and suppresses the H_2O_2 burst, finally favoring insect feeding.

showed a reduced NtRLP4 expression in *RNAi-NtRLP4* plants.

Immunohistochemistry (IHC) staining

Dissected *B. tabaci* heads with attached salivary glands and Parafilm-embedded salivary sheaths were fixed in 4% paraformaldehyde (#E672002, Sango Biotechnology, Shanghai) for 30 min. Salivary sheaths adherent to Parafilm were similarly excised and fixed under identical conditions. After fixation, samples were washed three times with PBST (PBS containing 0.1% Tween-20) and blocked with 10% FBS for 2 h at room temperature. The tissues were then incubated overnight at 4°C with Alexa Fluor 488-conjugated anti-BtRDP serum (1:200 dilution), followed by co-staining with phalloidin-rhodamine (1:500, Thermo Fisher Scientific) for 30 min and 4',6-diamidino-2-phenylindole (DAPI) solution (#ab104139, Abcam, Cambridge, USA) for nuclear visualization. Fluorescent images were acquired using a Leica SP8 confocal microscope (Leica Microsystems) to examine BtRDP localization patterns.

Protein extraction and Western-blotting assays

To identify secreted salivary proteins in *N. tabacum* plants, approximately 200 *B. tabaci* adults were confined in a 3-cm diameter feeding cage and allowed to feed on host plants for 24 h. Following feeding, the infested tobacco leaves were collected, immediately frozen in liquid nitrogen, and ground to a fine powder. The powdered tissue was then homogenized in RIPA Lysis Buffer (#89900, Thermo Fisher Scientific) for protein extraction. To examine the effects of insect feeding on NtRLP expression, groups of approximately 200 *B. tabaci* were allowed to feed on 4-5 true-leaf stage tobacco plants for 12, 24, or 48 h, with uninfested plants serving as controls. For silencing/overexpression efficiency tests, leaf samples were collected at 48 h post-*Agrobacterium* infiltration, along with transgenic tobacco plants at the 4-5 true-leaf stage. For insect tissue analysis, various *B. tabaci* organs were carefully dissected in 1×PBS buffer, including carcasses (20), fat bodies (20), guts (40), salivary glands (60), and ovaries (10), with sample sizes indicated in parentheses. All dissected tissues were transferred to lysis buffer for protein extraction. Total protein concentrations in both insect and plant homogenates were quantified using a BCA Protein Assay Kit (#CW0014S, CwBiotech, Taizhou, China). Equal protein amounts were separated by SDS-PAGE and electro-transferred to PVDF membranes.

Membranes were probed with the following primary antibodies: anti-FLAG (1:10,000, #MA1-91878, Thermo Fisher Scientific), anti-MYC (1:10,000, #MA1-21316, Thermo Fisher Scientific), anti-GFP (1:10,000, #MA5-15256, Thermo Fisher Scientific), anti-NtRLP4 (1:5,000, GenScript Biotechnology, Nanjing, China), and anti-BtRDP serum (1:5,000, Huaan Biotechnology, Hangzhou, China). Following primary antibody incubation, membranes were treated with HRP-conjugated secondary antibodies: goat anti-mouse IgG (1:10,000, #31430, Thermo Fisher Scientific) or goat anti-rabbit IgG (1:10,000, #31460, Thermo Fisher Scientific). Protein bands were visualized using an AI 680 image analyzer (Amersham Pharmacia Biotech, Buckinghamshire, UK), with Ponceau S staining used to verify equal protein loading.

Quantitative real-time PCR (qRT-PCR) analysis

To prepare tissue-specific samples from *B. tabaci*, adult females were dissected to isolate carcasses (10), fat bodies (10), guts (20), salivary glands (40), and ovaries (10). Additionally, different developmental stages, including eggs (50), nymphs (20), pupae (20), and adult males (20) and females (20), were collected for analysis. The number of insects in each sample is indicated in parentheses. To extract RNA from *N. tabacum* plants, the samples were grinded using liquid nitrogen after collection. Total RNA was extracted using the TRIzol Total RNA Isolation Kit (#9109, Takara, Dalian, China) following the manufacturer's instructions. RNA purity and concentration were assessed using a NanoDrop spectrophotometer (Thermo Fisher Scientific). First-strand cDNA was synthesized from equal amounts of RNA (1 µg) using the HiScript II Q RT SuperMix (#R212-01, Vazyme, Nanjing, China). Gene expression analysis was performed using a Roche LightCycler® 480

system (Roche Diagnostics, Mannheim, Germany) with SYBR Green Supermix (#11202ES08, Yeasen, Shanghai, China). The thermal cycling protocol consisted of initial denaturation at 95°C for 5 min, and amplification for 40 cycles of 95°C for 10 s and 60°C for 30 s. Primers were designed with Primer Premier v6.0 (Table S7). *B. tabaci Tubulin* and *N. tabacum Tubulin* served as internal controls for normalization. Relative gene expression was calculated using the $2^{-\Delta\Delta C_t}$ method [47]. Three to six independent biological replicates, each repeated twice, were performed.

RNA interference

The target gene DNA sequences were amplified using primers listed in Table S7 and subsequently cloned into the pClone007 Vector (#TSV-007, Tsingke Biotechnology, Beijing, China). For dsRNA production, PCR-generated DNA templates containing T7 promoter sequences were transcribed *in vitro* using the T7 High Yield RNA Transcription Kit (#TR101-01, Vazyme Biotech, Nanjing, China). The RNAi experiments were performed according to established methods [48]. Briefly, newly emerged adult whiteflies were anesthetized with carbon dioxide for 5-10 seconds. Approximately 50 nL of dsRNA solution (2 µg/µL) was microinjected into the mesothorax of each insect using a FemtoJet microinjection system (Eppendorf, Hamburg, Germany). Following injection, the whiteflies were maintained on healthy *N. tabacum* leaves under standard rearing conditions. Only actively moving insects were selected for subsequent analyses. Gene silencing efficiency was evaluated on day 4 post-injection using both qRT-PCR and Western-blotting approaches.

Generation of transgenic plants

We generated two transgenic *N. tabacum* lines expressing different BtRDP variants: one containing the full coding sequence (oeBtRDP) and another lacking the signal peptide (oeBtRDP^{SP}), which may affect protein secretion. The target sequences, along with the full-length NtRLP4 coding sequence and an NtRLP4 hairpin RNAi construct, were each cloned into the pBWA(V)HS vector with C-terminal FLAG tags. All constructs were transformed into *A. tumefaciens* strain GV3101 and used to generate transgenic tobacco plants (K326 background) through *Agrobacterium*-mediated transformation performed by Wuhan Boyuan Biological Company. Transgenic lines were verified by qRT-PCR and western blotting, with empty vector-transformed plants serving as controls, and two independent lines for each construct were selected for further experiments.

Target gene silencing in plants

The hairpin RNAi construct targeting *NtSOBIR1* was generated by cloning the specific fragment into the pBWA(V)HS plasmid using BsaI and Eco31I restriction enzymes. The recombinant vector was then transformed into *A. tumefaciens* strain GV3101 and subsequently infiltrated into *N. tabacum* leaves using the agroinfiltration method. Following a 48-h incubation period under standard growth conditions, silencing efficiency was quantitatively evaluated by qRT-PCR analysis.

Transcriptomic sequencing

To investigate the transcriptional effects of NtRLP4, BtRDP, and BtRDP^{SP} in *N. tabacum*, leaf samples from EV, oeRLP4#1, oeBtRDP#1, and oeBtRDP^{SP}#1 transgenic plants were homogenized in TRIzol Reagent (#10296018, Invitrogen, Carlsbad, CA, USA) for total RNA extraction following the manufacturer's protocol. Afterwards, total RNA was extracted according to the instructions of the manufacturer, and the RNA samples were sent to Novogene Institute (Novogene, Beijing, China) for transcriptomic sequencing. Briefly, poly(A)⁺ RNA was isolated from 20 µg pooled total RNA using oligo(dT) magnetic beads, followed by fragmentation in divalent cation buffer at 94°C for 5 min. First-strand cDNA synthesis was performed using N6 random primers, with subsequent second-strand synthesis to generate double-stranded cDNA. After end-repair and Illumina adapter ligation, the products were PCR-amplified (15 cycles) and purified using a QIAquick PCR Purification Kit (#28104, Qiagen, Hilden, Germany). The resulting cDNA libraries were quantified

and quality-checked before paired-end sequencing on an Illumina NovaSeq 6000 platform. All raw sequencing data were deposited in the National Genomics Data Center under accession number PRJCA025907.

Analysis of transcriptomic data

The raw sequencing reads were quality-filtered using Illumina's internal software, and the resulting clean reads from each cDNA library were aligned to the *N. tabacum* reference genome in Sol Genomics Network (https://solgenomics.net/ftp/genomes/Nicotiana_tabacum/edwards_et_al_2017/) using HISAT v2.1.0 [49]. Low-quality alignments were removed using SAMtools (v1.7) with default parameters [50]. Transcript abundance was quantified as Transcripts Per Million (TPM) using Cufflinks (v2.2.1) [51]. Differential expression analysis was performed with DESeq2 (v2.2.1) [52], identifying significantly DEGs using thresholds of $|\log_2(\text{fold change})| > 1$ and adjusted p -value < 0.05 . PCA of transcriptome profiles was conducted using a custom R script plotPCA (https://github.com/franco-ye/TestRepository/blob/main/PCA_by_deseq2.R). For functional annotation, KEGG pathway enrichment analysis was performed in TBtools (v2.083) using a one-sided hypergeometric test: $P = 1 - \sum_{i=0}^{m-1} \binom{M}{i} \binom{N-M}{n-i} / \binom{N}{n}$, [53]. In this software, enriched P -values were calculated according to one-sided hypergeometric test: with N representing the number of gene with KEGG annotation, n standing for the number of DEGs in N , M indicating the number of genes in each KEGG term, and m suggesting the number of DEGs in each KEGG term.

Insect bioassays

To survivorship analysis, newly emerged *B. tabaci* adults were microinjected with dsRNA and placed on *N. tabacum* leaves. After 24 h, any dead individuals (potentially due to injection injury) were removed. For each treatment, 20-30 surviving adults were transferred to leaf cages, and mortality was recorded daily for 10 days. Three independent biological replications were performed.

For fecundity analysis, dsRNA-treated virgin females were paired with untreated males in leaf cages. Five such pairs were maintained per cage ($n=10$ -12 cages per treatment). Egg deposition was quantified daily for 3 consecutive days under controlled conditions. Each male-female pair was considered an experimental replicate.

For host preference test, detached *N. tabacum* leaves with moist cotton-wrapped petioles were arranged in a 50-cm diameter choice arena with a central release chamber. Forty adult whiteflies were released per chamber, and settling preference was recorded at 3, 6, 12, 24, 36, and 48 h post-release. The experiment included 10-12 replicates per treatment, with leaf positions randomized between replicates to eliminate positional bias.

EPG recording and analysis

Feeding behavior of *B. tabaci* was monitored using a GiGA-8d EPG amplifier (Wageningen Agricultural University; 10 TΩ input resistance, <1 pA input bias current) with $50\times$ gain and ± 5 V output range. Prior to recording, dsRNA-treated whiteflies were starved for 12 h on moist filter paper, briefly anesthetized with CO₂ (10 s), and connected to the amplifier via a gold wire electrode (20 μm diameter $\times$ 5 cm length) attached to the pronotum using conductive silver glue. Plant electrodes consisted of copper wires (2 mm diameter $\times$ 10 cm length) inserted into the soil of potted *N. tabacum*. Recordings were conducted for 8 h in a Faraday cage (120 $\times$ 75 $\times$ 67 cm; Dianjiang, Shanghai), with only insects surviving the full duration retained for analysis.

Recorded waveforms were processed using PROBE 3.4 software (Wageningen Agricultural University) and categorized into four distinct feeding phases: nonpenetration (np), pathway duration (C), phloem salivation (E1), and phloem ingestion (E2) as defined in a previous study

[30]. Each treatment included ≥ 10 biological replicates, with dsGFP-injected and untreated whiteflies serving as controls.

Agrobacterium-mediated plant transformation in *N. tabacum* and *N. benthamiana*

The recombinant expression vectors were introduced into *A. tumefaciens* strain GV3101 via heat shock transformation. Transformed cells were selected on LB agar plates supplemented with 50 $\mu\text{g/mL}$ kanamycin and 10 $\mu\text{g/mL}$ rifampicin, followed by incubation at 28°C for 60 h. Positive colonies were inoculated into liquid LB medium with the same antibiotics and grown to log phase. Bacterial cells were then pelleted by centrifugation at 2,400 g for 2 min and resuspended in induction buffer (10 mM MgCl_2 , 10 mM MES [pH 5.6], 200 μM acetosyringone) adjusted to specific OD_{600} values depending on the experiment.

For standard experiments, bacterial suspensions were normalized to $\text{OD}_{600} = 1.0$, while degradation assays used three concentrations ($\text{OD}_{600} = 0.05, 0.3$, and 1.0). Equal volumes of selected bacterial suspensions were mixed and infiltrated into leaves of 4–5-week-old *N. tabacum* and *N. benthamiana* plants using a needleless syringe. Infiltrated plants were maintained under controlled conditions for subsequent analysis.

Diaminobenzidine (DAB) staining

To detect H_2O_2 accumulation in *N. tabacum* leaves, leaf samples were immersed in 1 mg/mL 3,3'-diaminobenzidine tetrahydrochloride (DAB-HCl, #D8001, Sigma-Aldrich, St. Louis, MO, USA) solution (pH 3.8) for 6 hours at room temperature, allowing H_2O_2 -dependent brown polymerization to occur. Following incubation, the DAB solution was replaced with 100% ethanol for chlorophyll removal, and samples were decolorized overnight at 65°C. The destained leaves were then photographed under standardized conditions using a Canon EOS 80D digital camera (Canon Inc., Tokyo, Japan).

Interaction assays between two proteins

In the Y2H screening assay, the coding sequence of BtRDP^{SP} was cloned into the pGBKT7 bait vector (Clontech, USA), while a *N. benthamiana* cDNA library was constructed in the pGADT7 prey vector (Biogene Biotech, Shanghai, China). These recombinant vectors were co-transformed into yeast strain Y2HGold and plated on QDO medium (SD/-Ade/-His/-Leu/-Trp, #630428, Takara) for 3 days at 30°C. Positive colonies were cultured in QDO liquid medium, and plasmids were extracted using a yeast DNA kit (#DP112-02, TIANGEN, Beijing, China) before transformation into *E. coli* DH5 α (#TSC-C14, Tsingke) for sequencing (YouKang Biotech, China) to identify interacting partners.

In the Y2H point-to-point verification assay, BtRDP^{SP} and NISP694^{SP}, and mutant variants of NtRPL4/SIRLP4/ OsRPL4 were cloned into either pGBKT7 or pGADT7 vectors using primers from Table S7. Yeast co-transformants were first selected on DDO medium (SD/-Leu/-Trp) (#630417, Takara) for 3 days at 30°C, then spotted onto QDO medium to confirm interactions through growth after another 3 days of incubation.

In the Co-IP assays, Full-length sequences of BtRDP, NISP694, NtSOBIR1, NtRPL4, SIRLP4, and OsRPL4 were cloned into lic-myc or lic-flag vectors for C-terminal tagging, with BtFTSP-flag serving as a negative control. Five-week-old *N. benthamiana* leaves were agroinfiltrated with these constructs or collected from oeBtRDP transgenic plants. Total proteins were extracted from 1 g frozen tissue in IP lysis buffer (#87788, Thermo Scientific) with protease inhibitors (#56079200, Roche, Switzerland), followed by centrifugation at 1,000 $\times g$ for 20 min. Cleared lysates were

incubated with anti-flag beads (#L00432-1, GenScript, Nanjing, China) for 4 h at 4°C, washed with PBS, and eluted in 2× SDS-PAGE buffer (500 mM Tris-HCl, pH=6.8, 50% glycerin, 10% SDS, 1% bromophenol blue, and 2% β-mercaptoethanol) for Western-blotting assay.

In the BiFC assay, BtRDP^{SP}, BtFTSP^{SP}, NtRLP4, and NtCf9-like were cloned into the pCV-cYFP or pCV-nYFP vectors, respectively. After transforming *A. tumefaciens* GV3101, pairwise combinations were co-infiltrated into *N. benthamiana* leaves. Following 36–48 h incubation in a growth chamber, reconstituted YFP fluorescence was visualized using a Leica SP8 confocal microscope with standard filter sets for YFP detection.

RLP degradation by salivary effectors among different herbivorous insects

To examine salivary protein effects on RLPs, we co-infiltrated *A. tumefaciens* GV3101 strains carrying various constructs into *N. benthamiana* leaves. The recombinant vectors expressed C-terminally tagged proteins: BtRDP-flag, NISP694-flag, GUS-flag, NISP7-flag, and myc-tagged RLPs (NtRLP4, SlRLP4, OsRLP4, SlCf9, NtCf9). Bacterial suspensions were mixed at predetermined OD₆₀₀ ratios before infiltration, with leaf samples collected 48 hours post-infiltration for subsequent analysis.

For confocal microscopy studies, *N. benthamiana* leaves were co-infiltrated with *A. tumefaciens* carrying: (1) NtRLP4-GFP plus BtRDP-mCherry, (2) NtRLP4-GFP plus RFP-mCherry, or (3) NtRLP4-GFP plus BtFTSP-mCherry. The samples were imaged by confocal microscopy at 48 h post injection. The same parameters were used in photographing.

This study also conducted a treatment on *N. benthamiana* plants that were overexpressing NtRLP4-myc using purified proteins. Briefly, the BtRDP^{SP} and GFP coding sequences were cloned into pET-28a for 6×His-tagged expression in *E. coli* Transetta (TransGen Biotech). Protein expression was induced with 0.5 mM IPTG at 25°C for 8 h, followed by purification using Ni-NTA agarose and buffer exchange into PBS using a 3-kDa molecular-weight cutoff Amicon Ultra-4 Centrifugal Filter Device (Millipore, MA, USA). Different concentrations of purified proteins were infiltrated into leaves of *N. benthamiana* plants stably expressing NtRLP4-myc, with samples collected 24 h post-treatment.

To investigate degradation pathways, we infiltrated *N. benthamiana* leaves with: 50 μM MG132 (#M7449, Sigma-Aldrich, Steinheim, Germany), 5 μM 3-MA (#M9281, Sigma-Aldrich), or 10 nM BAF (#BML-CM110-0100, Enzo Life Sciences, Farmingdale, USA) at 24 h post-agroinfiltration. The samples were incubated for additional 24 h. Protein levels were detected using Western-blotting assay.

In planta ubiquitination assay

Ubiquitination detection in *N. benthamiana* was performed according to the methods previously reported [54]. The recombinant vector for transient expression of the HA-fused ubiquitin protein was kindly provided by Dr. Chao Zheng (Ningbo University). Myc-tagged proteins were co-expressed with HA-UBQ by agroinfiltration, and 50 μM MG132 was injected into the *N. benthamiana* leaves one day post *A. tumefaciens* infiltration. Immunoprecipitation was performed using anti-myc beads (ytma-20, ChromoTek, Planegg-Martinsried, Germany). Ubiquitination signals were detected by immunoblot with anti-HA antibody (HT301, TransGen Biotech, Beijing, China).

Scanning electron microscopy (SEM)

B. tabaci adults were allowed to feed on artificial diet solutions for 24 h. Parafilm sections containing salivary sheaths were carefully excised and rinsed with 1× PBS (pH 7.4) to remove residual diet. The samples were mounted on aluminum stubs, desiccated under vacuum, and sputter-coated with gold before imaging with a TM4000 II Plus SEM (Hitachi, Tokyo, Japan). Salivary sheath length (from apex to base) was measured, with 20 sheaths analyzed per treatment to ensure statistical reliability.

Statistical analysis

The log-rank test (SPSS Statistics 19, Chicago, IL, USA) was used to determine the statistical significance of survival distributions. Two-tailed unpaired Student's *t*-test (comparisons between two groups) or one-way ANOVA test followed by Tukey's multiple comparisons test (comparisons among three groups) was used to analyze the results of qRT-PCR, fecundity analysis, and host choice analysis. The EPG data were first checked for the normality and homogeneity of variance, and those not fitting a normal distribution were $\log_{10}$ transformed before Student's *t* test, as described in previous studies [30]. Data were graphed in GraphPad Prism 9.

Data availability

All data is included in the manuscript and/or supporting information. The transcriptomic data have been submitted to the National Genomics Data Center under accession number PRJCA025907. The original blot images have been displayed in Fig. S30-S33.

Acknowledgements

This project has received funding from the National Key Research and Development Plan in the 14th five-year plan (2021YFD1401100: H.J.H, C.X.Z.), National Natural Science Foundation of China (32422075: H.J.H.; U23A6006: J.P.C and C.X.Z.), and the Natural Science Foundation of Zhejiang Province (LDQ24C140001: H.J.H).

Additional information

Author contributions

H-JH, J-ML, C-XZ, and J-PC planned and designed the research. XW, J-BL, Y-ZW, H-JH, H-BL, Z-LZ, and Z-XY performed experiments and analyzed data. X-HZ provided the EV plants. H-JH and XW drafted the manuscript.

Additional files

Supplementary file 1 [↗](#)

References

1. Nürnberger T, Brunner F, Kemmerling B, Piater L (2004) **Innate immunity in plants and animals: striking similarities and obvious differences** *Immunol Rev* **198**:249–266 [Google Scholar](#)
2. Macho AP, Zipfel C (2014) **Plant PRRs and the activation of innate immune signaling** *Mol Cell* **54**:263–272 [Google Scholar](#)
3. Zipfel C (2014) **Plant pattern-recognition receptors** *Trends Immunol* **35**:345–351 [Google Scholar](#)
4. Gust AA, Felix G (2014) **Receptor like proteins associate with SOBIR1-type of adaptors to form bimolecular receptor kinases** *Curr Opin Plant Biol* **21**:104–111 [Google Scholar](#)
5. Steinbrenner Adam D., et al. (2020) **A receptor-like protein mediates plant immune responses to herbivore-associated molecular patterns** *Proc Natl Acad Sci USA* **117**:31510–31518 [Google Scholar](#)
6. Chinchilla D, Bauer Z, Regenass M, Boller T, Felix G (2005) **The Arabidopsis receptor kinase FLS2 binds flg22 and determines the specificity of flagellin perception** *Plant Cell* **18**:465–476 [Google Scholar](#)
7. Zipfel C, et al. (2006) **Perception of the bacterial PAMP EF-Tu by the receptor EFR restricts Agrobacterium-mediated transformation** *Cell* **125**:749–760 [Google Scholar](#)
8. Ron M, Avni A (2004) **The receptor for the fungal elicitor ethylene-inducing xylanase is a member of a resistance-like gene family in tomato** *Plant Cell* **16**:1604–1615 [Google Scholar](#)
9. Tian T, et al. (2021) **A salivary calcium-binding protein from Laodelphax striatellus acts as an effector that suppresses defense in rice** *Pest Manag Sci* **77**:2272–2281 [Google Scholar](#)
10. Huang HJ, et al. (2019) **Salivary DNase II from Laodelphax striatellus acts as an effector that suppresses plant defence** *New Phytol* **224**:860–874 [Google Scholar](#)
11. Chen CY, et al. (2019) **An effector from cotton bollworm oral secretion impairs host plant defense signaling** *Proc Natl Acad Sci USA* **116**:14331–14338 [Google Scholar](#)
12. Xu HX, et al. (2019) **A salivary effector enables whitefly to feed on host plants by eliciting salicylic acid-signaling pathway** *Proc Natl Acad Sci USA* **116**:490–495 [Google Scholar](#)
13. Chen CY, Mao YB (2020) **Research advances in plant-insect molecular interaction** *F1000Res* **9**:198 [Google Scholar](#)
14. Guiguet A, et al. (2016) **Shared weapons of blood- and plant-feeding insects: Surprising commonalities for manipulating hosts** *J Insect Physiol* **84**:4–21 [Google Scholar](#)
15. Wu J, Baldwin IT (2009) **Herbivory-induced signalling in plants: perception and action** *Plant Cell Environ* **32**:1161–1174 [Google Scholar](#)

16. Jiang YJ, Zhang CX, Chen RZ, He SY (2019) **Challenging battles of plants with phloem-feeding insects and prokaryotic pathogens** *Proc Natl Acad Sci USA* **116**:23390–23397 [Google Scholar](#)
17. Gao H, et al. (2023) **The salivary chaperone protein NIDNAJB9 of Nilaparvata lugens activated plant immune responses** *J Exp Bot* **74**:6874–6888 [Google Scholar](#)
18. Guo H, et al. (2020) **An aphid-secreted salivary protease activates plant defense in phloem** *Curr Biol* **30**:4826–4836 [Google Scholar](#)
19. Bos JI, et al. (2010) **A functional genomics approach identifies candidate effectors from the aphid species Myzus persicae (green peach aphid)** *PLoS Genet* **6**:e1001216 [Google Scholar](#)
20. Zhang Y, Zeng L (2020) **Crosstalk between ubiquitination and other post-translational protein modifications in plant immunity** *Plant Commun* **1**:100041 [Google Scholar](#)
21. Huang J, et al. (2020) **Phytophthora effectors modulate genome-wide alternative splicing of host mRNAs to reprogram plant immunity** *Mol Plant* **13**:1470–1484 [Google Scholar](#)
22. Shan L, et al. (2008) **Bacterial effectors target the common signaling partner BAK1 to disrupt multiple MAMP receptor-signaling complexes and impede plant immunity** *Cell Host Microbe* **4**:17–27 [Google Scholar](#)
23. Huang HJ, et al. (2016) **Screening and functional analyses of Nilaparvata lugens salivary proteome** *J Proteome Res* **15**:1883–1896 [Google Scholar](#)
24. Huang HJ, et al. (2021) **Identification of salivary proteins in the whitefly Bemisia tabaci by transcriptomic and LC-MS/MS analyses** *Insect Sci* **28**:1369–1381 [Google Scholar](#)
25. Gao H, et al. (2022) **Nilaparvata lugens salivary protein NIG14 triggered defense response in plants** *J Exp Bot* **73**:7477–7487 [Google Scholar](#)
26. Mukhtar M, et al. (2011) **Independently evolved virulence effectors converge onto hubs in a plant immune system network** *Science* **333**:596–601 [Google Scholar](#)
27. Huang J, et al. (2020) **Salivary protein 1 of brown planthopper is required for survival and induces immunity response in plants** *Front Plant Sci* **11**:571280 [Google Scholar](#)
28. Wang Y, et al. (2023) **Horizontally transferred salivary protein promotes insect feeding by suppressing ferredoxin-mediated plant defenses** *Mol Biol Evol* **40**:msad221 [Google Scholar](#)
29. Huang HJ, et al. (2023) **Planthopper salivary sheath protein LsSP1 contributes to manipulation of rice plant defenses** *Nat Commun* **14**:737 [Google Scholar](#)
30. Lu S, et al. (2021) **EPG-recorded feeding behaviors reveal adaptability and competitiveness in two species of Bemisia tabaci (Hemiptera: Aleyrodidae)** *J Insect Behav* **34**:26–40 [Google Scholar](#)
31. Yan ZW, et al. (2023) **Endocytosis-mediated entry of a caterpillar effector into plants is countered by jasmonate** *Nat Commun* **14**:6551 [Google Scholar](#)
32. Jones DA, et al. (1994) **Isolation of the tomato Cf-9 gene for resistance to Cladosporium fulvum by transposon tagging** *Science* **266**:789–793 [Google Scholar](#)

33. Varshavsky A, et al. (2017) **The ubiquitin system, autophagy, and regulated protein degradation** *Annu Rev Biochem* **86**:123–128 [Google Scholar](#)
34. Li L, et al. (2022) **Independently evolved viral effectors convergently suppress DELLA protein SLR1-mediated broad-spectrum antiviral immunity in rice** *Nature Commun* **13**:6920 [Google Scholar](#)
35. Takatsuka C, et al. (2004) **3-Methyladenine inhibits autophagy in tobacco culture cells under sucrose starvation conditions** *Plant Cell Physiol* **45**:265–274 [Google Scholar](#)
36. Tariqjaveed M, et al. (2021) **Versatile effectors of phytopathogenic fungi target host immunity** *J Integr Plant Biol* **63**:1856–1873 [Google Scholar](#)
37. Doblaz VG, et al. (2018) **Cell wall integrity signaling in plants: Malectin-domain kinases and lessons from other kingdoms** *Cell Surf* **3**:1–11 [Google Scholar](#)
38. Li B, Lu D, Shan L (2014) **Ubiquitination of pattern recognition receptors in plant innate immunity** *Mol Plant Pathol* **15**:737–746 [Google Scholar](#)
39. Chen Y, Song Y, Liu J, Xu G, Dou D (2022) **Ubiquitination of receptorsomes, frontline of plant immunity** *Int J Mol Sci* **23**:2937 [Google Scholar](#)
40. Langin G, González-Fuente M, Üstün S (2023) **The plant ubiquitin–proteasome system as a target for microbial manipulation** *Annu Rev Phytopathol* **61**:351–375 [Google Scholar](#)
41. Rosas-Diaz T, et al. (2018) **A virus-targeted plant receptor-like kinase promotes cell-to-cell spread of RNAi** *Proc Natl Acad Sci USA* **115**:1388–1393 [Google Scholar](#)
42. Irieda H, et al. (2019) **Conserved fungal effector suppresses PAMP-triggered immunity by targeting plant immune kinases** *Proc Natl Acad Sci USA* **116**:496–505 [Google Scholar](#)
43. Qin J, et al. (2018) **The Xanthomonas effector XopK harbours E3 ubiquitin-ligase activity that is required for virulence** *New Phytol* **220**:219–231 [Google Scholar](#)
44. Xu FQ, Xue HW (2019) **The ubiquitin-proteasome system in plant responses to environments** *Plant Cell Environ* **42**:2931–2944 [Google Scholar](#)
45. Katoh K, Standley DM (2013) **MAFFT multiple sequence alignment software version 7: improvements in performance and usability** *Mol Biol Evol* **30**:772–780 [Google Scholar](#)
46. Kozlov AM, Darriba D, Flouri T, Morel B, Stamatakis A (2019) **RAxML-NG: a fast, scalable and user-friendly tool for maximum likelihood phylogenetic inference** *Bioinformatics* **35**:4453–4455 [Google Scholar](#)
47. Livak K.J., Schmittgen T.D (2001) **Analysis of relative gene expression data using real-time quantitative PCR and the 2– $\Delta\Delta$ CT method** *Methods* **25**:402–408 [Google Scholar](#)
48. Xu HJ, et al. (2015) **Two insulin receptors determine alternative wing morphs in planthoppers** *Nature* **519**:464–467 [Google Scholar](#)
49. Kim D., Langmead B., Salzberg S.L (2015) **HISAT: A fast spliced aligner with low memory requirements** *Nature Methods* **519**:464–467 [Google Scholar](#)

50. Li H (2009) **The Sequence Alignment/Map format and SAMtools** *Bioinformatics* [Google Scholar](#)
51. Trapnell C., Roberts A., Goff L., Pertea G., Kim D., Kelley D.R., Pimentel H., Salzberg S.L., Rinn J.L., Pachter L (2012) **Differential gene and transcript expression analysis of RNA-seq experiments with TopHat and Cufflinks** *Nature Protocols* **7**:562–578 [Google Scholar](#)
52. Wang L., Feng Z., Wang X., Wang X., Zhang X (2010) **DEGseq: an R package for identifying differentially expressed genes from RNA-seq data** *Bioinformatics* **26**:136–138 [Google Scholar](#)
53. Chen C., Chen H., Zhang Y., Thomas H.R., Frank M.H., He Y., Xia R (2020) **TBtools: an integrative toolkit developed for interactive analyses of big biological data** *Mol Plant* **13**:1194–1202 [Google Scholar](#)
54. Wang N., Yin Z., Zhao Y., Wang J., Pei Y., Ji P., Daly P., Li Z., Dou D., Wei L (2022) **An F-box protein attenuates fungal xylanase-triggered immunity by destabilizing LRR-RLP NbeIX2 in a SOBIR1-dependent manner** *New Phytol* **236**:2202–2215 [Google Scholar](#)

Author information

Xin Wang

State Key Laboratory for Quality and Safety of Agro-Products, Key Laboratory of Biotechnology in Plant Protection of MARA, Zhejiang Key Laboratory of Green Plant Protection, Institute of Plant Virology, Ningbo University, Ningbo, China, Institute of Applied Ecology, Fujian Agriculture and Forestry University, Fuzhou, China

Jia-Bao Lu

State Key Laboratory for Quality and Safety of Agro-Products, Key Laboratory of Biotechnology in Plant Protection of MARA, Zhejiang Key Laboratory of Green Plant Protection, Institute of Plant Virology, Ningbo University, Ningbo, China

Yi-Zhe Wang

State Key Laboratory for Quality and Safety of Agro-Products, Key Laboratory of Biotechnology in Plant Protection of MARA, Zhejiang Key Laboratory of Green Plant Protection, Institute of Plant Virology, Ningbo University, Ningbo, China

Xu-Hong Zhou

Office of Science and Technology, Yunnan University of Chinese Medicine, Kunming, China

Jian-Ping Chen

State Key Laboratory for Quality and Safety of Agro-Products, Key Laboratory of Biotechnology in Plant Protection of MARA, Zhejiang Key Laboratory of Green Plant Protection, Institute of Plant Virology, Ningbo University, Ningbo, China, Institute of Applied Ecology, Fujian Agriculture and Forestry University, Fuzhou, China

Chuan-Xi Zhang

State Key Laboratory for Quality and Safety of Agro-Products, Key Laboratory of Biotechnology in Plant Protection of MARA, Zhejiang Key Laboratory of Green Plant Protection, Institute of Plant Virology, Ningbo University, Ningbo, China
ORCID iD: [0000-0002-7784-1188](https://orcid.org/0000-0002-7784-1188)

Jun-Min Li

State Key Laboratory for Quality and Safety of Agro-Products, Key Laboratory of Biotechnology in Plant Protection of MARA, Zhejiang Key Laboratory of Green Plant Protection, Institute of Plant Virology, Ningbo University, Ningbo, China

For correspondence: lijunmin@nbu.edu.cn

Hai-Jian Huang

State Key Laboratory for Quality and Safety of Agro-Products, Key Laboratory of Biotechnology in Plant Protection of MARA, Zhejiang Key Laboratory of Green Plant Protection, Institute of Plant Virology, Ningbo University, Ningbo, China

ORCID iD: [0000-0002-0968-4520](https://orcid.org/0000-0002-0968-4520)

For correspondence: huanghaijian@nbu.edu.cn

Editors

Reviewing Editor

Kenichi Tsuda

Huazhong Agricultural University, Wuhan, China

Senior Editor

Sergio Rasmann

University of Neuchâtel, Neuchâtel, Switzerland

Reviewer #1 (Public review):

Summary:

This is a well-structured and interesting manuscript that investigates how herbivorous insects, specifically whiteflies and planthoppers, utilize salivary effectors to overcome plant immunity by targeting the RLP4 receptor.

Strengths:

The authors present a strong case for the independent evolution of these effectors and provide compelling evidence for their functional roles.

Weaknesses:

Western blot evidence for effector secretion is weak. The possibility of contamination from insect tissues during the sample preparation should be avoided.

Below are some specific comments and suggestions to strengthen the manuscript.

(1) Western blot evidence for effector secretion:

The western blot evidence in Figure 1, which aims to show that the insect protein is secreted into plants, is not fully convincing. The band of the expected size (~30 kDa) in the infested tissues is very weak. Furthermore, the high and low molecular weight bands that appear in the infested tissues do not match the size of the protein in the insects themselves, and a high molecular weight band also appears in the uninfested control tissues. It is difficult to draw a definitive conclusion that this protein is secreted into the plants based on this evidence. The authors should also address the possibility of contamination from insect tissues during the sample preparation and explain how they have excluded this possibility.

(2) Inconsistent conclusion (Line 156 and Figure 3c): T

The statement in line 156 is inconsistent with the data presented in Figure 3c. The figure clearly shows that the LRR domain of the protein is the one responsible for the interaction with BtRDP, not the region mentioned in the text. This is a critical misrepresentation of the experimental findings and must be corrected. The conclusion in the text should accurately reflect the data from the figure.

(3) Role of SOBIR1 in the RLP4/SOBIR1 Complex:

The authors demonstrate that the salivary effectors destabilize the RLP4 receptor, leading to a decrease in its protein levels and a reduction in the RLP4/SOBIR1 complex. A key question remains regarding the fate of SOBIR1 within this complex. The authors should clarify what happens to the SOBIR1 protein after the destabilization of RLP4. Does SOBIR1 become unbound, targeted for degradation itself, or does it simply lose its function without RLP4? This would provide further insight into the mechanism of action of the effectors.

(4) Clarification on specificity and evolutionary claims:

The paper's most significant claim is that the effectors from both whiteflies and planthoppers "independently evolved" to target RLP4. While the functional data is compelling, this evolutionary claim would be more convincing with stronger evidence. Showing that two different effector proteins target the same host protein is a fascinating finding but without a robust phylogenetic analysis, the claim of independent evolution is not fully supported. It would be valuable to provide a more detailed evolutionary analysis, such as a phylogenetic tree of the effector proteins, showing their relationship to other known insect proteins, to definitively rule out a shared, but highly divergent, common ancestor.

(5) Role of SOBIR1 in the interaction:

The results suggest that the effectors disrupt the RLP4/SOBIR1 complex. It is not entirely clear if the effectors are specifically targeting RLP4, SOBIR1, or both. Further experiments, such as a co-immunoprecipitation assay with just RLP4 and the effector, could clarify if the effector can bind to RLP4 in the absence of SOBIR1. This would help to definitively place RLP4 as the primary target.

(6) Transcriptome analysis (Lines 130-143):

The transcriptome analysis section feels disconnected from the rest of the manuscript. The findings, or lack thereof, from this analysis do not seem to be directly linked to the other major conclusions of the paper. This section could be removed to improve the manuscript's overall focus and flow. If the authors believe this data is critical, they should more clearly and explicitly connect the conclusions of the transcriptome analysis to the core findings about the effector-RLP4 interaction.

(7) Signal peptide experiments (Lines 145 and beyond):

The experiments conducted with the signal peptide (SP) are questionable. The SP is typically cleaved before the protein reaches its final destination. As such, conducting experiments with the SP attached to the protein may have produced biased observations and could lead to unjustified conclusions about the protein's function within the plant cell. We suggest the authors remove the experiments that include the signal peptide.

(8) Overly strong conclusion and unclear evidence (Line 176):

The use of the word "must" on line 176 is very strong and presents a definitive conclusion without sufficient evidence. The authors state that the proteins must interact with SOBIR1,

but they do not provide a clear justification for this claim. Is SOBIR1 the only interaction partner for NtRFP4? The authors should provide a specific reason for focusing on SOBIR1 instead of demonstrating an interaction with NtRFP4 first. Additionally, do BtRDP or NISP694 also interact with SOBIR1 directly? The authors should either tone down their language to reflect the evidence or provide a clearer justification for this strong claim.

<https://doi.org/10.7554/eLife.108737.1.sa2>

Reviewer #2 (Public review):

Summary:

The authors tested an interesting hypothesis that white flies and planthoppers independently evolved salivary proteins to dampen plant immunity by targeting a receptor-like protein.

Strengths:

The authors used a wide range of methods to dissect the function of the white fly protein BtRDP and identify its host target NtRFP4.

Weaknesses:

(1) Serious concerns about protein work.

I did not find the indicated protein bands for anti-BtRDP in Figures 1a and 1b in the original blot pictures shown in Figure S30. In Figure 1a, I can't get the point of showing an unspecific protein band with a size of ~190 kD as a loading control for a protein of ~30 kD.

The data discrepancy led me to check other Western blot pictures. Similarly, Figures 2d, 3b, 3d, and S15b (anti-Myc) do not correspond to the original blots shown. In addition, the anti-Myc blot in Figure 4i, all blot pictures in Figures 5b, 5h, and S19a appeared to be compressed vertically. These data raised concerns about the quality of the manuscript.

Blots shown in Figure 3d, 4f, 4g, and 4h appeared to be done at a different exposure rate compared to the complete blot shown in Figure S30. The undesirable connection between Western blot pictures shown in the figures and the original data might be due to the reduced quality of compressed figures during submission. Nevertheless, clarification will be necessary to support the strength of the data provided.

(2) Misinterpretation of data.

I am afraid the authors misunderstood pattern-triggered immunity through receptor-like proteins. It is true that several LRR-type RLPs constitutively associate with SOBIR1, and further recruit BAK1 or other SERKs upon ligand binding. One should not take it for granted that every RLP works this way. To test the hypothesis that NtRFP4 confers resistance to *B. tabaci* infestation, the author compared transcriptional profiles between an EV plant line and an RFP4 overexpression line. If I understood the methods and figure legends correctly, this was done without *B. tabaci* treatment. This experimental design is seriously flawed. To provide convincing genetic evidence, independent mutant lines (optionally independent overexpression lines) in combination with different treatments will be necessary. Otherwise, one can only conclude that overexpressing the RFP4 protein generated a nervous plant. In addition, ROS burst, but not H₂O₂ accumulation, is a common immune response in pattern-triggered immunity.

(3) Lack of logic coherence.

The written language needs substantial improvement. This impeded the readability of the work. More importantly, the logic throughout the manuscript appeared scattered. The choice of testing protein domains for protein-protein interactions, using plants overexpressing an insect protein to study its subcellular localization, switching back and forth between using proteins with signal peptides and without signal peptides, among others, lacks a clear explanation.

<https://doi.org/10.7554/eLife.108737.1.sa1>

Reviewer #3 (Public review):

Summary:

In this study, Wang et al. investigate how herbivorous insects overcome plant receptor-mediated immunity by targeting plant receptor-like proteins. The authors identify two independently evolved salivary effectors, BtRDP in whiteflies and NISP694 in brown planthoppers, that promote the degradation of plant RLP4 through the ubiquitin-dependent proteasome pathway. NtRLP4 from tobacco and OsRLP4 from rice are shown to confer resistance against herbivores by activating defense signaling, while BtRDP and NISP694 suppress these defenses by destabilizing RLP4 proteins.

Strengths:

This work highlights a convergent evolutionary strategy in distinct insect lineages and advances our understanding of insect-plant coevolution at the molecular level.

Weaknesses:

- (1) I found the naming of BtRDP and NISP694 somewhat confusing. The authors defined BtRDP as "B. tabaci RLP-degrading protein," whereas NISP694 appears to have been named after the last three digits of its GenBank accession number (MF278694, presumably). Is there a standard convention for naming newly identified proteins, for example, based on functional motifs or sequence characteristics? As it stands, the inconsistency makes it difficult for readers to clearly distinguish these proteins from those reported in other studies.
- (2) Figure 2 and other figures. Transgenic experiments require at least two independent lines, because results from a single line may be confounded by position effects or unintended genomic alterations, and multiple lines provide stronger evidence for reproducibility and reliability.
- (3) Figure 3e. Quantitative analysis of NtRLP4 was required. Additionally, since only one band was observed in oeRLP, were any tags included in the construct?
- (4) Figure 4a. The RNAi effect appears to be well rescued in Line 1 but poorly in Line 2. Could the authors clarify the reason for this difference?
- (5) ROS accumulation is shown for only a single leaf. A quantitative analysis of ROS accumulation across multiple samples would be necessary to support the conclusion. The same applies to Figure 16f.
- (6) Figure 4f: NtRLP4 abundance was significantly reduced in oeBtRDP plants but not in oeBtRDP-SP. Although coexpression analysis suggests that BtRDP promotes NtRLP4 degradation in an ubiquitin-dependent manner, the reduced NtRLP4 levels may not result from a direct interaction between BtRDP and NtRLP4. It is possible that BtRDP influences other factors that indirectly affect NtRLP4 abundance. The authors should discuss this possibility.

(7) The statement in lines 335-336 that 'Overexpression of NtRLP4 or NtSOBIR1 enhances insect feeding, while silencing of either gene exerts the opposite effect' is not supported by the results shown in Figures S16-S19. The authors should revise this description to accurately reflect the data.

(8) BtRDP is reported to attach to the salivary sheath. Does the planthopper NISP694 exhibit a similar secretion localization (e.g., attachment to the salivary sheath)? The authors should supplement this information or discuss the potential implications of any differences in secretion localization between BtRDP and NISP694 for their respective modes of action.

<https://doi.org/10.7554/eLife.108737.1.sa0>